

UNIFIED BI, REPORTING & AI ANALYTICS PLATFORM

Functional boards, reporting capabilities and AI-driven insights

A consolidated reference for product, business, UX, architecture, development, testing and operational stakeholders.

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TICVAI • Powered by Intelligence

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Board 1 — Unified Analytics Command Center

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide executives, senior management, venue managers, and authorized operational users with a **single real-time command center** covering the overall performance of the TICVAI ecosystem. The dashboard shall consolidate information from Ticketing, POS, B2B, Access Control, Finance, F&B, Retail, Membership, Loyalty, CRM, Resource Management, Virtual Queue, and other connected modules.



The board shall support **multi-site, multi-venue, multi-business-unit and multi-currency operations**, with role-based visibility, real-time KPI monitoring, drill-down analysis, historical comparison, AI-generated insights, and configurable alerts.

The underlying matrix specifically requires a real-time management overview including revenue, site/product/channel performance, visitors, capacity utilization, conversion, average spend, repeat customers, membership statistics, receivables, refunds, and top-performing products/channels.

Screen 1 — Executive Command Center

Purpose

Provide senior management with an immediate accurate consolidated view of overall organizational performance.

Functional Scope

The system shall display:

- Total Revenue — Today / MTD / YTD
- Revenue vs Target
- Revenue vs Previous Period

- Total Transactions
- Tickets Sold
- Total Visitors
- Current Occupancy
- Capacity Utilization %
- Conversion Rate
- Average Spend per Visitor
- Refund Value & %
- Active Members
- Repeat Customer %
- Outstanding Receivables
- Top Performing Site
- Top Performing Product
- Top Performing Sales Channel

The screen shall support configurable KPI cards, charts and widgets.

Global Filters

- Organization / Tenant
- Country
- Site / Venue
- Attraction
- Business Unit
- Date / Period
- Sales Channel
- Product Category
- Currency

AI Capability

An **AI Executive Summary** shall automatically explain major changes, exceptions and opportunities.

Example:

“Revenue is 8.4% above target today, primarily driven by online ticket sales and higher weekend attendance. F&B spend per visitor is 6.2% below the four-week average.”

Acceptance Criteria

Management can view consolidated business performance and navigate from enterprise-level KPIs to supporting detail without accessing individual operational modules.

Screen 2 — Live Operations Dashboard

Purpose

Provide a real-time view of what is happening across all operating locations.

Functional Scope

Display:

- Visitors currently on-site

- Entries today
- Exits today
- Current occupancy
- Capacity remaining
- Occupancy %
- Active gates
- Gate status
- Active POS terminals
- Active sessions/timeslots
- Current queues
- Resource utilization
- Operational incidents
- System/device exceptions

Sites shall be visually classified as:

Normal / Warning / Critical

The requirements specifically call for operational reporting covering active licenses, workstations, POS, users and user activity.

AI Capability

AI shall prioritize operational issues based on business impact.

Example:

“Gate 4 rejection rate has increased to 11% during the last 15 minutes. Primary reason: expired credential validation.”

Acceptance Criteria

Authorized operations users can identify the live status of each site and immediately drill into abnormal conditions.

Screen 3 — Revenue Pulse

Purpose

Provide real-time visibility into revenue generation across TICVAI.

Functional Scope

Display:

- Revenue Today
- Revenue This Hour
- Revenue vs Target
- Revenue vs Yesterday
- Revenue vs Same Day Last Week
- Revenue vs Same Period Last Year
- Revenue by Site
- Revenue by Attraction
- Revenue by Product
- Revenue by Channel
- Revenue by Business Unit

- Revenue by Payment Method

Business-unit breakdown shall include, where applicable:

Ticketing | F&B | Retail | Membership | B2B | Add-ons | Parking | Other Services

The matrix requires daily, weekly and monthly revenue reporting, including revenue by attraction/location and before/after allocations.

AI Capability

AI shall identify major revenue drivers and negative variances.

“Revenue is AED 126K below forecast. 61% of the variance is associated with lower admission volume at Site B.”

Acceptance Criteria

Management can identify revenue performance and drill down from consolidated revenue to site → business unit → channel → product → transaction level, subject to permissions.

Screen 4 — Attendance & Footfall Intelligence

Purpose

Monitor visitor movement and attendance across venues.

Functional Scope

Display:

- Total Visitors
- Current Visitors On-Site
- Entry Count
- Exit Count
- Hourly Footfall
- Peak Entry Time
- Average Visit Duration
- Attendance by Site
- Attendance by Attraction
- Attendance by Ticket Type
- Attendance by Product
- Attendance by Timeslot
- Repeat Visits
- No-Show %
- Ticketed vs Actual Attendance

Visualizations shall include an hourly attendance curve and site/attraction comparison.

The reporting requirements explicitly include people count, ticket count by gate/site, admission usage, attendance by hour/day/product and occupancy reporting.

AI Capability

AI shall identify abnormal attendance patterns and forecast expected attendance.

Acceptance Criteria

Users can understand historical, current and forecast visitor volumes and drill into the underlying attendance data.

Screen 5 — Capacity & Utilization Monitor

Purpose

Provide centralized monitoring of available and consumed capacity.

Functional Scope

Display utilization for:

- Venue
- Attraction
- Session
- Timeslot
- Resource
- Seating area
- Virtual Queue
- Gate
- Operational resource

KPIs:

- Available Capacity
- Booked Capacity
- Used Capacity
- Remaining Capacity
- Utilization %
- No-Show %
- Peak Utilization
- Forecast Utilization

The source requirements also call for available-versus-booked capacity, peak utilization, session occupancy and resource utilization reporting.

Alerts

Threshold examples:

- 80% — Warning
- 95% — Critical
- 100% — Full

Thresholds shall be configurable rather than hard-coded.

Acceptance Criteria

Users can identify underutilized, approaching-capacity and fully utilized locations/resources in real time.

Screen 6 — Sales & Channel Performance

Purpose

Provide management with consolidated commercial performance across every sales channel.

Channels

- B2C Web
- Mobile App

- POS
- Kiosk
- B2B
- Reseller
- OTA
- Call Center
- Corporate
- Partner/API

KPIs

- Gross Sales
- Net Sales
- Transactions
- Tickets Sold
- Average Order Value
- Conversion Rate
- Discount Value
- Refunds
- Cancellations
- Commission
- Upsell Revenue
- Cross-Sell Revenue

Analytics

Allow comparison by:

Channel × Site × Product × Period × Customer Segment.

The matrix includes detailed sales-channel reporting covering ticket types, serial numbers, quantity, targets, commissions, refunds, cancellations and discounts.

AI Capability

AI shall identify high/low-performing channels and unusual changes.

Acceptance Criteria

Management can compare channel profitability and sales performance from one dashboard.

Screen 7 — Customer, Membership & Loyalty Pulse

Purpose

Provide a consolidated view of customer health and engagement.

Functional Scope

Display:

- Unique Customers
- New Customers
- Returning Customers
- Repeat Visit %

- Average Customer Spend
- Customer Lifetime Value
- Active Members
- New Memberships
- Membership Renewals
- Membership Expiring
- Loyalty Members
- Loyalty Earn
- Loyalty Burn
- Outstanding Loyalty Liability
- Customer Satisfaction Score

Segmentation

Allow filtering by:

- Customer segment
- Membership type
- Geography
- Demographic group
- Acquisition source
- Visit frequency
- Spend tier

The source matrix requires customer demographics, acquisition, repeat-customer analysis, lifetime value, spending trends, loyalty usage and segmentation.

AI Capability

AI shall identify retention opportunities and unusual customer behavior.

Acceptance Criteria

Authorized users can understand customer acquisition, engagement, retention, membership and loyalty performance without accessing separate CRM dashboards.

Screen 8 — Alerts & Exception Center

Purpose

Create one centralized location for management-level KPI and operational exceptions.

Alert Categories

- Revenue
- Sales
- Finance
- Access Control
- Capacity
- Ticketing
- Payment
- Membership
- Loyalty

- F&B
- Retail
- Inventory
- System/Integration

Each Alert Shall Display

- Severity
- KPI
- Current value
- Expected/target value
- Variance
- Site
- Detection time
- Source system
- Owner
- **Status**
- Recommended action

Alert Workflow

New → Acknowledged → Assigned → Investigating → Resolved

The reporting matrix explicitly calls for exceptional reports when KPIs fall outside expected ranges.

AI Capability

AI shall:

- Explain probable cause
- Assess business impact
- Correlate related alerts
- Recommend actions
- Prioritize alerts

Acceptance Criteria

Management can identify and manage important business exceptions from a centralized queue.

Screen 9 — AI Management Insights

Purpose

Provide management with proactive AI-generated business intelligence rather than requiring users to manually analyze every dashboard.

Insight Types

Performance

- Revenue movements
- Attendance movements
- Channel performance

Anomalies

- Unexpected refund increase

- Gate rejection spikes
- Revenue deviation
- Unusual discount usage

Opportunities

- Upsell opportunity
- Cross-sell opportunity
- Capacity optimization
- Membership conversion
- Dynamic pricing opportunity

Forecasts

- Expected attendance
- Expected revenue
- Capacity demand
- Membership renewals

AI Insight Structure

Each insight should show:

Observation → Cause → Impact → Recommendation → Confidence → Supporting Data

Example:

Observation: Saturday 14:00–16:00 sessions are consistently above 92% utilization.

Impact: Estimated AED 48K potential demand is being lost monthly.

Recommendation: Evaluate additional session capacity or dynamic pricing.

Governance

AI recommendations shall be advisory unless an explicitly configured approval workflow allows execution.

Acceptance Criteria

Users receive explainable, traceable AI insights linked to the underlying supporting data.

Screen 10 — Multi-Site & Performance Comparison

Purpose

Allow enterprise management to compare sites, venues, attractions and business units.

Comparison KPIs

- Revenue
- Revenue Growth
- Attendance
- Capacity Utilization
- Revenue per Visitor
- Average Transaction Value
- Conversion
- Refund %
- Membership Conversion
- Repeat Visitor %
- Customer Satisfaction

- F&B Spend
- Retail Spend
- Operational Exceptions

Benchmarking

Users shall compare:

Site vs Site

Venue vs Venue

Attraction vs Attraction

Actual vs Target

Current vs Previous Period

Current vs Previous Year

Ranking functionality shall identify:

- Top performers
- Lowest performers
- Largest improvement
- Largest decline
- Outliers

AI Capability

AI shall automatically explain meaningful performance differences.

“Site A generates 14% more revenue per visitor than Site B, primarily due to higher F&B attachment and membership penetration.”

Acceptance Criteria

Management can benchmark operating entities using normalized KPIs and drill into the causes of performance differences.

Board-Wide Functional Requirements

All 10 screens shall support, where applicable:

Real-Time Refresh — Live KPIs shall refresh automatically according to configurable refresh intervals.

Historical Analytics — Users shall move between live, daily, weekly, monthly, quarterly and yearly analysis.

Drill-Down / Drill-Through — KPI → chart → site → product/channel → supporting records.

Role-Based Security — Data shall respect tenant, organization, site, business unit, operating area and user permissions.

Multi-Site — Authorized users can consolidate multiple sites.

Multi-Currency — Local and reporting/base currency shall be supported.

Multi-Language — English and Arabic shall be supported; the broader reporting specification explicitly requires both.

Export — Authorized dashboards/data shall support Excel, PDF, CSV and/or API export according to permissions.

Auditability — Dashboard access, sensitive drill-down actions, exports and relevant configuration changes shall be logged.

Embedded BI — Analytics shall appear as an integrated part of TICVAI rather than requiring normal operational users to leave TICVAI for a separate BI portal.

AI Principle for Board 1

One important design rule should be included in the SOW:

Every major management dashboard shall contain an AI Insight layer capable of explaining what happened, identifying likely contributing factors, forecasting what may happen next, and recommending appropriate management actions.

This turns Board 1 from a conventional **Power BI dashboard** into the actual **TICVAI AI-powered Management Command Center**.

Board 1 Screen Summary

#	Screen
1.1	Executive Command Center
1.2	Live Operations Dashboard
1.3	Revenue Pulse
1.4	Attendance & Footfall Intelligence
1.5	Capacity & Utilization Monitor
1.6	Sales & Channel Performance
1.7	Customer, Membership & Loyalty Pulse
1.8	Alerts & Exception Center
1.9	AI Management Insights
1.10	Multi-Site & Performance Comparison

This Board 1 should be designed as the **“wow” landing screen of TICVAI management**: when a CEO, GM, CFO, or Operations Director logs in, this is the area that immediately tells them **what is happening, why it is happening, what is likely to happen next, and where they need to act**.

Board 2 — Dashboard Designer & KPI Studio

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide authorized TICVAI administrators, analysts, and business owners with a **no-code / low-code workspace to create, configure, govern, preview, and publish dashboards and KPIs** without requiring software development for every new reporting requirement.



Board 1 was the **consumption layer**—management sees what is happening. Board 2 is the **configuration layer**—authorized users decide **what dashboards contain, how KPIs are calculated, which data is used, who can see them, and how they behave**.

This directly supports the requirement for a self-service reporting platform, dashboard designer, role-based access, multi-site reporting, interactive dashboards, KPI scorecards, drill-down/drill-through, trend analysis and forecasting.

Screen 2.1 — Dashboard Library

Purpose

Provide a centralized catalogue for all standard, custom, AI-generated and embedded TICVAI dashboards.

Functional Scope

Display dashboard cards/table containing:

- Dashboard Name
- Dashboard ID
- Category
- Business Domain
- Owner
- Sites/Venues
- Audience/Role

- **Status**
- Version
- Last Modified
- Last Published
- Usage Count
- Data Refresh Status

Dashboard Categories

- Executive
- Operations
- Finance
- Sales
- Ticketing
- Access Control
- CRM
- Marketing
- Membership
- Loyalty
- F&B
- Retail
- Inventory
- Resources
- Custom

Dashboard Types

System Dashboard — TICVAI standard dashboard

Custom Dashboard — customer-created

AI-Generated Dashboard — generated through AI

Embedded Dashboard — displayed inside another TICVAI module

Actions

Create • Clone • Edit • Preview • Publish • Unpublish • Archive • Export Definition • View Usage • View History

Acceptance Criteria

Authorized users can search, filter, manage, clone and govern dashboards from one central catalogue.

Screen 2.2 — Dashboard Creation Wizard

Purpose

Guide users through creation of a new dashboard.

Creation Steps

Step 1 — Basic Information

Configure:

- Dashboard Name
- Description
- Category
- Business Domain

- Dashboard Owner
- Tags
- Language
- Default Currency

Step 2 — Scope

Select:

- Tenant
- Organization
- Site(s)
- Venue(s)
- Attraction(s)
- Business Unit(s)

Step 3 — Data Domains

Select one or multiple domains:

Ticketing • Sales • Access • Finance • Customer • CRM • Membership • Loyalty • F&B • Retail • Inventory • Resource • Queue • Accreditation

Step 4 — Template

Choose:

- Blank Dashboard
- TICVAI Template
- Existing Dashboard
- AI-Generated Starting Point

Step 5 — Audience

Select roles/groups permitted to access the dashboard.

AI Capability

User may enter:

“Create an operations dashboard for the General Manager showing today's revenue, visitors, capacity, gate performance and customer spend.”

AI proposes an initial dashboard structure.

Acceptance Criteria

Users can create the initial dashboard configuration without technical/database knowledge.

Screen 2.3 — Drag-and-Drop Dashboard Canvas

Purpose

Provide the main visual workspace for dashboard construction.

Functional Scope

The canvas shall support:

- Drag-and-drop widgets
- Resize
- Move
- Align
- Snap-to-grid

- Duplicate
- Delete
- Group
- Lock position
- Full-width sections
- **Tabs**
- Containers
- Responsive layout

Layout Options

Desktop • Tablet • Mobile • Large Operations Display

Canvas Controls

Undo • Redo • Zoom • Grid • Preview • Save Draft • Validate • Publish

Widget Properties

Selecting any widget opens a configuration panel for:

- Data
- Visualization
- **Filters**
- Formatting
- Interaction
- Drill-down
- Permissions
- Refresh behavior

Acceptance Criteria

A business analyst can visually construct a dashboard without coding.

Screen 2.4 — Widget & Visualization Library

Purpose

Provide reusable visual components for dashboard construction.

KPI Components

- KPI Card
- Target Card
- Variance Card
- Scorecard
- Gauge
- Progress Indicator

Chart Components

- Line
- Bar
- Column
- Area
- Pie/Donut

- Scatter
- Waterfall
- Funnel
- Heatmap
- Treemap

Operational Components

- Table
- Matrix
- Map
- Site Status
- Gate Status
- Occupancy
- Capacity
- Queue Monitor
- Alert Feed

AI Components

- AI Executive Summary
- AI Insight Card
- Forecast
- Anomaly Indicator
- Recommendation Card
- Natural-Language Query Box

Acceptance Criteria

Authorized users can search the component library and drag supported visualizations directly onto the dashboard canvas.

Screen 2.5 — KPI Builder

Purpose

Allow authorized business users to create standardized enterprise KPIs.

KPI Definition

Configure:

- KPI Name
- KPI Code
- Description
- Business Domain
- Owner
- Data Source
- Measure
- Formula
- Aggregation
- Unit
- Currency

- Decimal Precision
- Effective Date
- **Status**

Example

KPI: Revenue per Visitor

Formula:

Net Revenue ÷ Admitted Visitors

Comparison Configuration

Compare against:

- Target
- Budget
- Yesterday
- Previous Week
- Previous Month
- Previous Year
- Same Day Last Year
- Custom Period

Direction

Define whether:

Higher = Better

or

Lower = Better

This is important for KPIs such as:

Revenue ↑

Conversion ↑

Refund Rate ↓

Gate Rejection Rate ↓

Queue Time ↓

Acceptance Criteria

Administrators can centrally define reusable KPIs and ensure the same KPI calculation is used across TICVAI dashboards.

Screen 2.6 — Targets, Thresholds & KPI Status Rules

Purpose

Configure how TICVAI determines whether KPI performance is healthy, warning or critical.

Configuration

For each KPI:

- Target
- Minimum
- Maximum
- Warning Threshold
- Critical Threshold
- Benchmark

- Tolerance
- Evaluation Frequency

Example

Capacity Utilization

Target: 80%

Normal: 0–80%

Warning: 81–94%

Critical: ≥95%

Conditional Rules

Rules may differ by:

- Site
- Venue
- Attraction
- Business Unit
- Product
- Day
- Season
- Event
- Operating period

Alert Connection

Users can configure:

If Critical → Generate Alert

and optionally:

- Notify responsible user
- Email
- Mobile notification
- Create operational task
- Trigger AI analysis

The source requirements specifically call for exceptional reports when defined KPIs fall outside acceptable ranges.

Acceptance Criteria

KPI status and alert behavior are calculated dynamically using centrally configured business thresholds.

Screen 2.7 — Data & Filter Configuration

Purpose

Control what data a dashboard/widget uses and how users can filter it.

Data Configuration

Select:

- Data Domain
- Dataset
- Measure
- Dimension
- Aggregation

- Date Field
- Relationship
- Calculation

Common Dimensions

- Site
- Venue
- Attraction
- Product
- Ticket Type
- Channel
- Customer Segment
- Membership Type
- Salesperson
- POS
- Gate
- Business Unit
- Payment Type

Dashboard Filters

Configure:

- Date Range
- Site
- Venue
- Attraction
- Channel
- Product
- Customer Segment
- Currency

Filter Behavior

Filters may be:

- Global
- Page-level
- Widget-level
- Mandatory
- Optional
- Hidden
- Defaulted

Cross-Domain Analytics

Users shall be able to combine authorized datasets.

Example:

Ticket Sales + Attendance + F&B Spend + Membership

This supports the requirement for cross-domain reporting rather than isolated reporting silos.

Acceptance Criteria

Users can configure dashboard data and filters without direct access to underlying production databases.

Screen 2.8 — Drill-Down & Interaction Designer

Purpose

Configure how users move from high-level KPIs into deeper analytics.

Example Drill Hierarchy

Total Revenue

↓

Site

↓

Business Unit

↓

Channel

↓

Product

↓

Transaction

Another example:

Attendance

↓

Venue

↓

Attraction

↓

Gate

↓

Ticket Type

↓

Admission Record

Interaction Configuration

Define:

- Click behavior
- Cross-filtering
- Drill-down
- Drill-up
- Drill-through
- Tooltip
- Detail page
- Related dashboard
- Underlying report

Security

Drill-down shall never bypass the user's data permissions.

A user authorized only for **Site A** cannot drill into Site B through a consolidated widget.

The reporting platform requirements explicitly require both drill-down and drill-through capability.

Acceptance Criteria

Dashboard designers can visually configure analytical navigation without custom development.

Screen 2.9 — Dashboard Access, Publishing & Versioning

Purpose

Govern who can access dashboards and how dashboard changes reach production.

Access Configuration

Assign dashboards to:

- Users
- Roles
- Groups
- Departments
- Sites
- Operating Areas
- Business Units

Permission Levels

View

Interact

Export

Share

Edit

Administer

The existing requirements explicitly require report access based on group rights, operating-area rights and individual user access rights.

Lifecycle

Draft → Review → Approved → Published → Superseded → Archived

Versioning

Display:

- Version Number
- Changed By
- Date/Time
- Change Description
- Approval Status
- Published Version

Rollback

Administrators shall be able to restore a previous approved dashboard version.

Acceptance Criteria

Only authorized and approved dashboard configurations become available to production users, with complete version history.

Screen 2.10 — Dashboard Preview, Validation & Health

Purpose

Validate dashboards before publication and monitor their technical/analytical health afterward.

Preview Modes

- Desktop
- Tablet
- Mobile
- Operations Display
- Executive View

Validation Checks

TICVAI shall automatically check:

- Missing data sources
- Broken KPIs
- Invalid formulas
- Broken filters
- Broken drill paths
- Permission conflicts
- Unsupported visualizations
- Slow queries
- Missing translations
- Stale data
- Failed refreshes

Dashboard Health

Display:

- Data Last Refreshed
- Refresh Frequency
- Dataset Status
- Query Performance
- Widget Load Time
- Failed Widgets
- API Status
- User Count
- Usage Frequency

AI Optimization

AI may recommend:

“This dashboard contains three widgets querying the same dataset. Consolidating the queries may improve dashboard loading performance.”

or:

“The Revenue by Channel widget has not been viewed for 90 days. Consider removing or replacing it.”

Publish Readiness

Final status:

Ready to Publish

or

Issues Detected

with blocking and non-blocking issues clearly identified.

Acceptance Criteria

The system prevents dashboards containing critical configuration or security errors from being published.

Board-Wide AI Capabilities

AI shall assist throughout the dashboard configuration lifecycle.

Users should be able to type:

“Build a CFO dashboard.”

“Add a chart comparing revenue against last year.”

“Show me which KPI is missing from this operations dashboard.”

“Create an attendance forecast for the next seven days.”

“Explain why this dashboard is loading slowly.”

AI can **recommend and configure**, but governed changes shall continue to respect approval, security and publication controls.

Important Design Principle — Power BI Integration

Board 2 should **not look like the Power BI administration portal**.

The end user should experience:

TICVAI Dashboard Designer

while the underlying analytics engine may be:

Power BI / Microsoft Fabric / TICVAI Analytics Services / another approved enterprise BI technology.

This allows us to replace or extend the underlying BI technology later without redesigning the TICVAI user experience.

Board 2 Screen Summary

#	Screen	Main Purpose
2.1	Dashboard Library	Manage all dashboards
2.2	Dashboard Creation Wizard	Create/configure dashboards
2.3	Drag-and-Drop Dashboard Canvas	Visually build dashboards
2.4	Widget & Visualization Library	Select charts, KPIs and AI widgets
2.5	KPI Builder	Define enterprise KPIs
2.6	Targets, Thresholds & KPI Status Rules	Configure targets and exceptions
2.7	Data & Filter Configuration	Map data and filters
2.8	Drill-Down & Interaction Designer	Configure analytical navigation
2.9	Access, Publishing & Versioning	Govern dashboard lifecycle
2.10	Preview, Validation & Health	Test and monitor dashboards

Board 2 Core Flow

Dashboard Library → Create → Select Data → Build Canvas → Add Widgets → Define KPIs → Configure Filters → Configure Drill-Down → Set Permissions → Validate → Approve → Publish

This board gives TICVAI something very important: **Soft Labs does not need to develop a new screen every time a client asks for a new management dashboard.** The client can create governed dashboards themselves, while TICVAI controls the data, KPI definitions, security, AI layer, approval and publishing

Board 3 — Self-Service Report Builder

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide authorized TICVAI users with a powerful **no-code self-service reporting environment** where they can create, run, customize, save, and manage reports without requesting development work or requiring knowledge of databases, SQL, Power BI, or the underlying technical architecture.

While **Board 2** focuses on building visual dashboards and KPIs, **Board 3** focuses on creating detailed operational and analytical **reports**: tables, matrices, transaction listings, summaries, reconciliations, cross-domain reports, and ad-hoc analysis.

The source matrix explicitly requires customized reports based on available fields and records, as well as configurable reporting across the platform. It also requires a self-service report builder with multi-site reporting, drill-down/drill-through, interactive analytics, and multiple export formats.

Screen 3.1 — Report Catalogue & Library

Purpose

Provide a centralized repository containing all TICVAI standard, custom, scheduled and AI-generated reports.

Report Categories

- Executive
- Ticketing
- Sales
- Reservations
- Access & Admissions
- Finance
- Payments
- B2B

- CRM & Customers
- Membership
- Loyalty
- Marketing
- F&B
- Retail
- Inventory
- Resources
- Operations
- Audit & Security

Report Information

Each report shall display:

- Report Name
- Report ID
- Description
- Category
- Business Domain
- Owner
- Report Type
- Site Scope
- Created Date
- Last Run
- Last Modified
- **Status**
- Usage Count

Report Types

System Report — TICVAI standard report

Custom Report — created by authorized users

AI-Generated Report — created using natural language

Shared Report — shared by another user

Scheduled Report — configured for automatic execution

Actions

Run • Edit • Clone • Share • Schedule • Export • Archive • View History

Acceptance Criteria

Users can search, filter, organize and access authorized reports from one centralized report library.

Screen 3.2 — Report Creation Wizard

Purpose

Guide users through creation of a new report without requiring technical knowledge.

Step 1 — Report Information

Configure:

- Report Name

- Description
- Category
- Owner
- Tags
- Language

Step 2 — Business Domain

Select one or multiple domains:

Ticketing • Sales • Reservations • Finance • Payments • Access • Customers • Membership • Loyalty • F&B • Retail • Inventory • Resources • Marketing

Step 3 — Report Type

Choose:

- Detailed Report
- Summary Report
- Transaction Report
- Matrix Report
- Comparison Report
- Reconciliation Report
- Audit Report
- Exception Report

Step 4 — Starting Point

Choose:

- Blank
- TICVAI Template
- Existing Report
- AI-Generated

AI Creation

Users may type:

“Create a daily sales report showing revenue by site, attraction, product and sales channel, including discounts and refunds.”

AI shall propose the required fields, filters, grouping and calculations.

Acceptance Criteria

An authorized business user can create the initial structure of a report without knowing the underlying database schema.

Screen 3.3 — Data Domain & Dataset Selector

Purpose

Allow users to select approved TICVAI data sources for reporting.

Available Domains

Examples:

Ticketing

Orders • Tickets • Products • Reservations • Timeslots • Ticket Usage

Sales

Transactions • Channels • Discounts • Promotions • Upsell/Cross-Sell

Finance

Revenue • Tax • Receivables • Revenue Recognition • Cost Centers

Payments

Payments • Refunds • Tender • Reconciliation

Access

Admissions • Gates • Entries • Exits • Rejections

Customer

Profiles • Visits • Segments • Spend

Membership/Loyalty

Memberships • Renewals • Earn/Burn • Redemptions

F&B / Retail

Sales • Products • Cost of Sales • Wastage • Inventory

Dataset Information

Show:

- Dataset Name
- Description
- Domain
- Available Date Range
- Refresh Frequency
- Last Refresh
- Data Owner
- Security Classification

Acceptance Criteria

Only approved semantic datasets shall be exposed to report creators; direct production-database access shall not be required.

Screen 3.4 — Field & Column Selector**Purpose**

Allow users to visually select which information appears in the report.

Available Fields

Fields shall be grouped logically.

Example for Sales:

Transaction

- Transaction ID
- Order ID
- Date
- Time
- **Status**

Location

- Site
- Venue
- Attraction
- POS

Product

- Product
- Ticket Type
- Category

Financial

- Gross Amount
- Discount
- Tax
- Refund
- Net Revenue

Channel

- Sales Channel
- Partner
- Reseller

Customer

- Customer ID
- Segment
- Membership Type

User Actions

Drag Field → Selected Columns

Users can:

- Add/remove columns
- Reorder columns
- Rename display labels
- Hide columns
- Set width
- Configure formatting
- Define sorting

Acceptance Criteria

Users can construct the report output using approved business-friendly field names.

Screen 3.5 — Filter & Parameter Builder

Purpose

Allow report creators to define which records are included.

Standard Filters

- Date Range
- Site
- Venue
- Attraction
- Business Unit
- Product
- Ticket Type

- Channel
- Customer
- Membership
- POS
- Cashier
- Payment Method
- Transaction Status

The existing requirements specifically require transaction searching by date range, sales station/workstation, time, cashier, client, email, amount, status, order/PNR and channel.

Operators

- Equals
- Not Equals
- Contains
- Begins With
- Greater Than
- Less Than
- Between
- In List
- Is Empty
- Is Not Empty

Filter Logic

Support:

AND / OR / Nested Conditions

Example:

Site = Adventure Park

AND

Channel = Online OR POS

AND

Net Revenue > AED 500

Parameters

Report creators may allow users to enter parameters when running the report.

Example:

Select Site: All Sites

Select Period: Last 30 Days

Acceptance Criteria

Users can create complex report filters without SQL or technical query syntax.

Screen 3.6 — Grouping, Aggregation & Calculation Builder

Purpose

Turn detailed transactional data into meaningful management reporting.

Grouping

Users can group by:

- Site
- Attraction
- Product
- Ticket Type
- Channel
- Customer Segment
- Cashier
- Date
- Week
- Month
- Year

Aggregations

Support:

- Sum
- Count
- Distinct Count
- Average
- Minimum
- Maximum
- Percentage

Calculated Fields

Authorized users can create formulas such as:

Net Revenue

Gross Revenue - Discounts - Refunds

Average Spend

Net Revenue / Visitors

Conversion Rate

Completed Transactions / Qualified Sessions × 100

Running Calculations

Support:

- Running Total
- Percentage of Total
- Variance
- Growth %
- Period-over-Period Change
- Ranking

AI Formula Assistant

Users may ask:

“Create a calculated field showing refund percentage.”

AI proposes:

Refund Value / Gross Sales × 100

The user must be able to review the proposed calculation before saving it.

Acceptance Criteria

Business users can create summaries and derived metrics using governed calculation tools.

Screen 3.7 — Cross-Domain Report Composer

Purpose

Allow authorized users to combine information across multiple TICVAI modules.

This is one of the most important capabilities of Board 3.

Example 1

Ticketing + Access

Tickets Sold

vs

Tickets Actually Used

Example 2

Attendance + F&B + Retail

Visitors

vs

F&B Spend

vs

Retail Spend

Example 3

CRM + Membership + Sales

Customer Segment

vs

Membership Status

vs

Lifetime Spend

Example 4

Sales + Finance + Payment

Ticketing Revenue

vs

Payment Gateway Settlement

vs

ERP Posting

Relationship Configuration

The platform shall use governed relationships between datasets rather than requiring users to manually create database joins.

Relationship Preview

Example:

Order → Ticket → Admission → Customer → Payment

Data Validation

TICVAI shall identify:

- Invalid relationships
- Duplicate aggregation risk
- Missing relationships
- Incompatible datasets
- Security restrictions

AI Assistance

AI may recommend:

“To calculate spend per admitted visitor, combine the Admission dataset with Sales using Site, Visit Date and Customer/Transaction relationship.”

Acceptance Criteria

Authorized users can create cross-domain analytics while maintaining governed data relationships and security.

Screen 3.8 — Report Layout & Formatting Designer

Purpose

Control how the final report is displayed.

Layout Components

- Report Header
- Title
- Logo
- Filter Summary
- Table
- Matrix
- Totals
- Subtotals
- Chart
- Footer
- Page Number
- Generated Date/Time

Formatting

Configure:

- Column labels
- Number format
- Currency
- Percentage
- Date format
- Decimal precision
- Conditional formatting
- Page orientation
- Page size

Conditional Formatting Example

Refund Rate:

Normal

Warning

Critical

Branding

Support:

- TICVAI branding
- Client branding

- Site branding
- Report logo
- Header/footer templates

Language

Support report output in:

English / Arabic

including RTL presentation where applicable.

Acceptance Criteria

Users can produce professional reports suitable for operational, financial and management use.

Screen 3.9 — Report Preview, Test & Validation

Purpose

Allow users to verify report accuracy before saving or publishing it.

Preview

Display:

- Sample Results
- Total Records
- Applied Filters
- Calculated Fields
- Grouping
- Totals
- Execution Time
- Data Refresh Time

Validation Checks

TICVAI shall check:

- Invalid field
- Broken relationship
- Invalid formula
- Missing parameter
- Security violation
- Duplicate aggregation
- Excessive data volume
- Query timeout risk
- Stale dataset
- Missing translation

AI Validation

AI may highlight:

“Revenue may be double-counted because Order Revenue is being combined with multiple Ticket records.”

This is particularly important for financial reporting.

Performance Recommendation

Example:

“This report retrieves 8.4 million transaction records. Consider applying a date filter before execution.”

Acceptance Criteria

Critical data, security or calculation issues must be identified before the report is published.

Screen 3.10 — Save, Run & Report Results Viewer

Purpose

Provide the final operational interface for executing and consuming reports.

Report Header

Display:

- Report Name
- Description
- Owner
- Last Run
- Data Refresh
- Applied Parameters
- Reporting Period

Runtime Controls

Users can:

Run Report

Refresh

Change Parameters

Drill Down

Drill Through

Sort

Search

Expand/Collapse

Export

Authorized users can export to:

- Excel
- PDF
- CSV
- XML

The matrix requires reporting export to multiple formats including PDF, Excel, delimited text and XML.

API-based report/data access shall also be supported where enabled.

Save Options

- Save
- Save As
- Save Personal View
- Save Shared View
- Add to Favorites
- Add to Dashboard

Further Actions

- Share

- Schedule
- Subscribe

The detailed configuration for scheduling and subscriptions will be handled in **Board 4**.

Acceptance Criteria

Users can execute reports interactively, analyze the results and export authorized data in supported formats.

AI — “Ask TICVAI to Build My Report”

Board 3 should include AI throughout the report-building process rather than making AI a separate disconnected feature.

A user could type:

“Show me all online ticket sales for Adventure Park during August, grouped by product and channel, with quantity, gross revenue, discounts, refunds and net revenue. Compare it against August last year.”

TICVAI AI should identify:

Domain: Sales + Ticketing

Period: August

Site: Adventure Park

Channel: Online

Grouping: Product + Channel

Measures: Quantity, Gross Revenue, Discount, Refund, Net Revenue

Comparison: Same month previous year

Then generate the report configuration for the user to review.

The AI must **not invent data or bypass permissions**. It creates a governed analytical query using TICVAI's approved semantic model.

Important Distinction Between Boards 2 and 3

We should keep this clear for Soft Labs.

Board 2	Board 3
Dashboard Designer	Report Builder
Visual management dashboards	Detailed reporting
KPI cards/charts	Rows, columns, matrices
Executive/operational monitoring	Analysis/reconciliation/detail
Persistent dashboard layout	Parameter-driven report execution
Visualization focused	Data/report focused

They share the same governed analytics/data layer but solve different user requirements.

Board 3 Screen Summary

#	Screen	Main Purpose
3.1	Report Catalogue & Library	Central report repository
3.2	Report Creation Wizard	Guided report creation
3.3	Data Domain & Dataset Selector	Select governed datasets
3.4	Field & Column Selector	Build report columns

#	Screen	Main Purpose
3.5	Filter & Parameter Builder	Configure report conditions
3.6	Grouping, Aggregation & Calculation Builder	Create summaries/calculations
3.7	Cross-Domain Report Composer	Combine TICVAI domains
3.8	Report Layout & Formatting Designer	Format final report
3.9	Report Preview, Test & Validation	Validate accuracy/security
3.10	Report Results Viewer	Run, analyze and export

Board 3 Core Flow

Report Library → Create Report → Select Data → Select Fields → Configure Filters → Group & Calculate → Combine Domains → Format → Validate → Run / Save

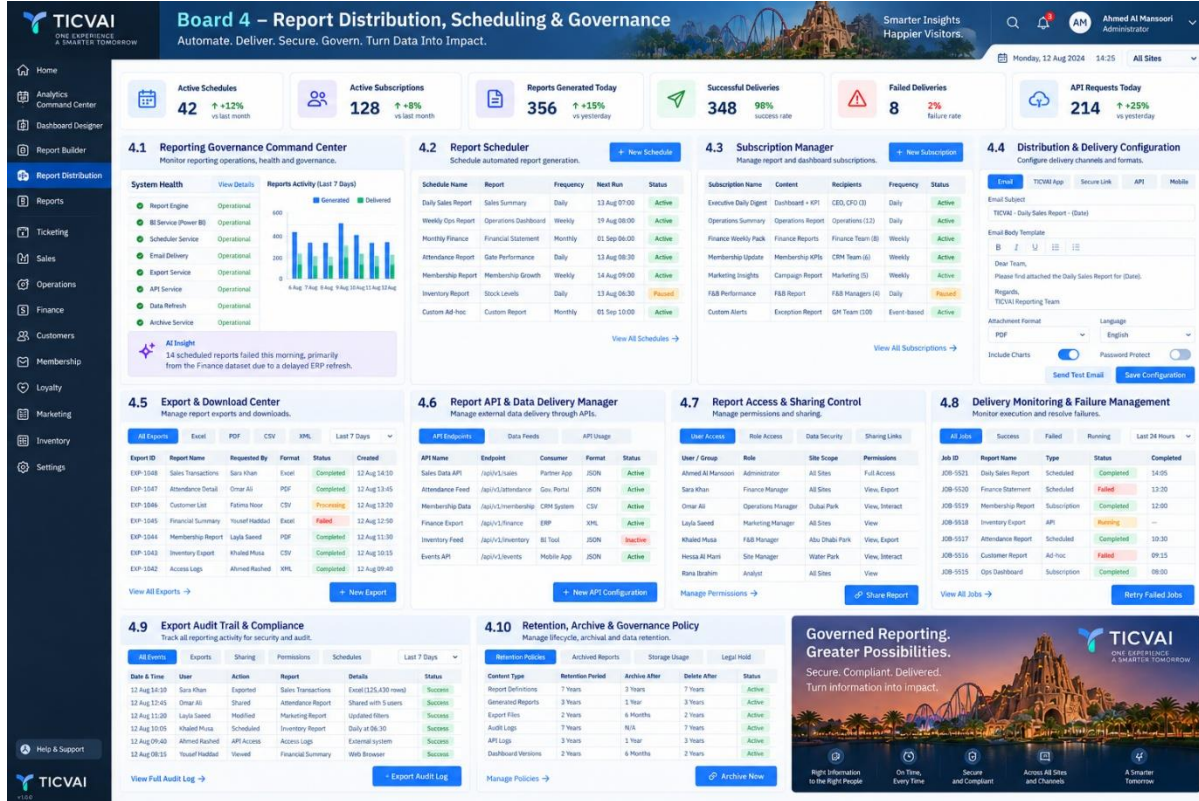
This board is what makes the requirement “**self-service reporting**” genuinely **self-service** rather than simply giving the client access to a collection of prebuilt Power BI reports.

Board 4 — Report Distribution, Scheduling & Governance

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide TICVAI with a centralized **Report Distribution & Governance Center** to control what happens after a dashboard or report has been created: **who receives it, when it runs, how it is delivered, what format is used, who may share/export it, how failures are handled, and how every action is audited.**



This board complements:

Board 1: Consume dashboards and insights

Board 2: Design dashboards and KPIs

Board 3: Build reports

Board 4: Schedule → Subscribe → Distribute → Share → Export → Secure → Audit → Retain

The source requirements explicitly call for scheduling report generation and delivery, report logs, and role-/operating-area-/user-based report access. The broader reporting platform also requires scheduled generation, subscriptions, role-based access, multi-site reporting, multiple export formats, API access, and long-term archival/retrieval.

Screen 4.1 — Reporting Governance Command Center

Purpose

Provide administrators with a centralized overview of reporting operations, governance, distribution and compliance.

KPI Summary

Display:

- Active Reports
- Published Dashboards
- Active Schedules
- Active Subscriptions
- Reports Generated Today

- Successful Deliveries
- Failed Deliveries
- Pending Approvals
- Exports Today
- API Report Requests
- Shared Reports
- Governance Exceptions

Operational Status

Show health by:

- Report Engine
- BI/Power BI Service
- Scheduler
- Email Delivery
- Export Service
- API Service
- Data Refresh
- Archive Service

Recent Activity

Display:

- Recently generated reports
- Failed reports
- Recently shared reports
- Large exports
- Permission changes
- Scheduled-report changes

AI Capability

AI shall summarize important reporting issues.

Example:

“14 scheduled reports failed this morning. Eleven failures originate from the Finance dataset following a delayed ERP refresh.”

Acceptance Criteria

Administrators can understand overall reporting-platform health and governance from one screen.

Screen 4.2 — Report Scheduler

Purpose

Allow authorized users to automatically generate reports according to defined schedules.

Schedule Configuration

Configure:

- Schedule Name
- Report
- Report Version

- Owner
- Start Date
- End Date
- Time
- Time Zone
- Frequency

Frequency Options

- Once
- Hourly
- Daily
- Weekly
- Monthly
- Quarterly
- Yearly
- Custom

Business Scheduling

Support examples such as:

Daily Sales Report

Every day at 07:00

Weekly Operations Report

Every Monday at 08:00

Monthly Finance Report

First business day of each month

Report Parameters

Each schedule can store:

- Site
- Venue
- Business Unit
- Period
- Currency
- Channel
- Product
- Other report parameters

Runtime Behavior

Configure:

- Run immediately
- Retry on failure
- Maximum retries
- Timeout
- Skip if source data is stale
- Wait for source refresh

Acceptance Criteria

Authorized users can create and manage recurring report-generation schedules without technical support.

Screen 4.3 — Subscription Manager

Purpose

Allow users and administrators to subscribe recipients to dashboards, reports and KPI summaries.

Subscription Types

- Full Report
- Dashboard Snapshot
- KPI Summary
- Exception Report
- AI Executive Summary
- Operational Digest

Recipient Types

- Individual User
- User Group
- Role
- Department
- Site Management
- Distribution List

Frequency

Users can select:

- Daily
- Weekly
- Monthly
- Event-based
- Exception-only

Example

Executive Daily Digest

Recipients:

CEO • CFO • COO

Content:

Revenue
Attendance
Capacity
Refunds
Major Alerts
AI Summary

Delivery:

Every day at 08:00

User Preferences

Where permitted, users may:

- Subscribe

- Unsubscribe
- Pause
- Change frequency
- Select format

Acceptance Criteria

Authorized users can manage report subscriptions without modifying the original report definition.

Screen 4.4 — Distribution & Delivery Configuration

Purpose

Configure how generated reporting content reaches approved recipients or destinations.

Delivery Channels

Support configurable channels such as:

- TICVAI Notification Center
- Email
- Secure Download
- API
- External approved destination
- Mobile notification for KPI/alert summaries

Delivery Configuration

Define:

- Destination
- Recipient
- Format
- Language
- Time Zone
- File Naming
- Password/Security Policy
- Expiration
- Delivery Priority

Email Delivery

Configure:

- Email Subject
- Email Body Template
- Attachment
- Secure Link
- Language
- Branding

Secure-Link Delivery

Links may support:

- Expiry Date
- Maximum Downloads

- Authentication Required
- Recipient Restriction

Security Principle

Sensitive financial/customer reports should not automatically be distributed as unrestricted attachments where organizational security policies require controlled access.

Acceptance Criteria

TICVAI can securely distribute reporting output through configured and approved delivery channels.

Screen 4.5 — Export & Download Center

Purpose

Provide centralized management of report and dashboard exports.

The source matrix requires reporting export to multiple formats, including PDF, Excel, delimited text and XML.

Supported Formats

At minimum:

- Excel
- PDF
- CSV
- XML

Where configured:

- JSON
- API Dataset

Export Information

Display:

- Export ID
- Report
- Requested By
- Site
- Format
- Requested Time
- Completed Time
- File Size
- Record Count
- **Status**
- Expiration

Export Status

Queued → Processing → Completed → Downloaded → Expired

or

Failed

Large Exports

Large datasets may be processed asynchronously.

Example:

“Your export contains 4.8 million records and will be processed in the background.”

Security Controls

Configure:

- Export Permission
- Maximum Rows
- Sensitive Field Masking
- File Expiry
- Encryption
- Download Limits

Acceptance Criteria

Users can export authorized information while TICVAI maintains control and auditability over data extraction.

Screen 4.6 — Report API & Data Delivery Manager**Purpose**

Allow approved systems to receive reporting information programmatically.

This supports the requirement for report export/access through APIs.

API Configuration

Configure:

- API Name
- Report/Dataset
- Consumer/Application
- Authentication
- Allowed Sites
- Allowed Fields
- **Filters**
- Rate Limit
- Data Format
- **Status**

Delivery Modes

Support:

On-Demand API

External system requests authorized reporting data.

Scheduled Data Feed

TICVAI generates reporting data according to schedule.

API Monitoring

Display:

- Requests Today
- Success Rate
- Failed Requests
- Response Time
- Data Volume
- Last Request

- Consumer

Security

API reporting must respect:

- Tenant boundaries
- Data permissions
- Field restrictions
- Authentication
- Rate limits
- Audit logging

Acceptance Criteria

External approved systems can consume governed reporting data without receiving unrestricted access to TICVAI's operational database.

Screen 4.7 — Report Access & Sharing Control

Purpose

Provide centralized security for reports, dashboards and analytical information.

The matrix specifically requires reporting access based on group rights, operating-area rights and user access rights.

Permission Assignment

Assign by:

- User
- Role
- Group
- Department
- Site
- Venue
- Operating Area
- Business Unit

Permission Levels

Configure:

View

Run

Interact

Drill Down

Export

Share

Schedule

Subscribe

Edit

Administer

Data-Level Security

Example:

Dubai Operations Manager

Can see:

Dubai Sites

Cannot see:

Saudi Arabia Sites

Sensitive Information

Specific fields may require additional authorization:

- Customer PII
- Financial Details
- Payment Information
- Staff Information
- Audit Information

Sharing

Users may share:

- Report
- Dashboard
- Saved View
- Secure Link

but only within their permitted security scope.

Acceptance Criteria

Sharing, export and subscription cannot be used to bypass the underlying TICVAI permission model.

Screen 4.8 — Delivery Monitoring & Failure Management

Purpose

Monitor scheduled report execution and distribution and provide operational management of failures.

Monitoring Table

Display:

- Job ID
- Report
- Schedule
- Start Time
- Completion Time
- Recipient Count
- Delivery Method
- **Status**
- Error
- Retry Count

Status

- Running
- Completed
- Partially Delivered
- Failed

- Cancelled
- Waiting for Data

Failure Reasons

Examples:

- Dataset unavailable
- Data refresh incomplete
- Report timeout
- Invalid recipient
- Email delivery failure
- API destination unavailable
- Permission changed
- Export generation failure

Actions

- Retry
- Cancel
- Re-run
- View Error
- Change Destination
- Notify Owner
- Escalate

AI Root-Cause Assistance

Example:

“Finance reports are failing because the ERP dataset has not completed its scheduled refresh. Recommendation: hold report execution until data refresh succeeds.”

Acceptance Criteria

Administrators can identify and resolve report-generation and distribution failures without reviewing technical server logs.

Screen 4.9 — Report Audit Trail & Compliance

Purpose

Maintain complete traceability of reporting activity.

The source matrix requires report logs and audit reporting for sales history.

Audit Events

Capture:

- Report Created
- Report Modified
- Report Published
- Report Executed
- Report Viewed
- Report Exported
- Report Shared
- Schedule Created

- Schedule Modified
- Subscription Created
- Permission Changed
- API Access
- Report Archived
- Report Deleted

Audit Record

Store:

- User
- Action
- Date/Time
- Report
- Version
- Site/Tenant
- Source
- Destination
- Result
- Reason where applicable

Search

Filter audit history by:

- User
- Report
- Action
- Date
- Site
- Business Unit
- Export Type
- **Status**

High-Risk Activity

Flag events such as:

- Very large export
- Sensitive customer-data export
- Unusual download activity
- Permission escalation
- Repeated failed API access

AI Capability

AI can identify suspicious reporting behavior for review.

Acceptance Criteria

Authorized administrators and auditors can reconstruct the history of relevant reporting activity.

Screen 4.10 — Retention, Archive & Governance Policy

Purpose

Control the lifecycle of reports, generated files and historical analytical data.

The reporting-platform specification requires long-term reporting history together with archival and retrieval capability.

Retention Policies

Configure retention separately for:

- Report Definitions
- Dashboard Versions
- Generated Reports
- Export Files
- Audit Logs
- Schedule History
- API Logs
- Historical Analytics

Policy Configuration

Define:

- Retention Period
- Archive After
- Delete After
- Legal/Compliance Hold
- Storage Location
- Retrieval Rules

Historical Reporting

The source matrix includes historical-record requirements for audit purposes.

The architecture should therefore support client-specific retention policies rather than hard-code one duration globally.

Archive Search

Users with permission can locate archived reports by:

- Report
- Period
- Site
- Owner
- Category
- Archive Date

Restore

Authorized users may:

Locate → Preview Metadata → Request/Restore → View/Export

Acceptance Criteria

Historical reporting content can be governed, archived and retrieved according to configured organizational and regulatory policies.

Board-Wide AI Capabilities

AI should support governance and distribution, but **must not override security policies**.

AI can:

- Detect failed schedules
- Explain delivery failures
- Identify unusual exports
- Recommend schedule optimization
- Detect duplicate report subscriptions
- Identify unused scheduled reports
- Recommend retention/archive actions
- Summarize governance exceptions

Example:

AI Governance Insight

17 weekly reports are being distributed to overlapping Finance recipient groups. Consolidating them into 6 subscriptions could reduce duplicate report generation by approximately 40%.

Important Approval Concept

For sensitive reporting, we should support an optional approval workflow.

For example:

User requests export of customer-level financial data

↓

TICVAI identifies Sensitive Data

↓

Approval Required

↓

Finance/Data Owner approves

↓

Export generated

↓

Audit trail created

This should be **configurable**, not mandatory for every export.

Board 4 Screen Summary

#	Screen	Main Purpose
4.1	Reporting Governance Command Center	Overall reporting governance and health
4.2	Report Scheduler	Automate report generation
4.3	Subscription Manager	Manage recurring report subscriptions
4.4	Distribution & Delivery Configuration	Configure delivery channels
4.5	Export & Download Center	Govern Excel/PDF/CSV/XML exports
4.6	Report API & Data Delivery Manager	External reporting/data delivery
4.7	Report Access & Sharing Control	Security, permissions and sharing
4.8	Delivery Monitoring & Failure Management	Monitor/report failures
4.9	Report Audit Trail & Compliance	Full reporting audit history

#	Screen	Main Purpose
4.10	Retention, Archive & Governance Policy	Historical reporting governance

Board 4 Core Flow

Create Report → Publish → Assign Access → Schedule / Subscribe → Generate → Distribute / API / Export → Monitor → Audit → Archive

Relationship Across the First Four Boards

Board 1 — “What is happening?” Unified Analytics Command Center.

Board 2 — “What should my dashboard show?” Dashboard Designer & KPI Studio.

Board 3 — “What detailed information do I need?” Self-Service Report Builder.

Board 4 — “Who gets it, when, how, and under what controls?” Report Distribution, Scheduling & Governance.

This separation is important because it prevents Soft Labs from building reporting as a collection of disconnected Power BI screens. Instead, TICVAI gets a **complete enterprise reporting platform with creation, consumption, distribution, security, governance and AI intelligence.**

Board 5 — Sales, Ticketing & Revenue Analytics

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide commercial, ticketing, revenue, and management teams with a unified analytics workspace for understanding **what was sold, where it was sold, through which channel, at what price, with which discounts/promotions, how much was refunded, how much revenue was generated, and how actual performance compares with targets and**

historical periods.



This board is the **commercial intelligence layer**. It shall consume governed data from Ticketing, Pricing, Promotions, POS, B2C, B2B, Resellers/OTA, Payments, CRM and Finance without duplicating the operational screens of those modules.

The source requirements specifically include reporting for daily transactions, historical sales, discounts, refunds, sales channels, promotions, commissions, upsell/cross-sell, revenue allocation, ticket usage and detailed sales performance.

Screen 5.1 — Sales & Revenue Command Center

Purpose

Provide a consolidated management view of overall commercial performance.

KPI Cards

Display:

- Gross Sales
- Net Sales
- Net Revenue
- Transactions
- Tickets Sold
- Average Order Value
- Average Ticket Price
- Revenue per Visitor
- Discounts
- Refunds
- Cancellations
- Complimentary Tickets
- Conversion Rate

- Revenue vs Target
- Revenue Growth %

Revenue Trend

Support comparison against:

- Yesterday
- Previous Week
- Previous Month
- Previous Year
- Budget
- Target
- Forecast

Revenue Breakdown

Analyze by:

Site → Attraction → Business Unit → Product → Channel

AI Executive Summary

Example:

“Net revenue is 7.8% above target today. Online sales contributed 46% of revenue, while B2B sales declined 11% compared with the same weekday last week.”

Acceptance Criteria

Management can determine overall commercial performance from a single screen and drill into the underlying drivers.

Screen 5.2 — Ticket Sales Analytics

Purpose

Provide detailed analytics specifically for ticket sales and ticket-product performance.

KPI Summary

- Tickets Sold
- Gross Ticket Revenue
- Net Ticket Revenue
- Average Ticket Price
- Paid Tickets
- Complimentary Tickets
- Discounted Tickets
- Cancelled Tickets
- Refunded Tickets
- Used Tickets
- Unused Tickets
- Expired Tickets

The source requirements specifically require reporting on sold admissions and whether tickets are used, unused, redeemed or expired, together with related revenue recognition.

Analysis Dimensions

Analyze by:

- Ticket Type

- Product
- Event
- Attraction
- Timeslot
- Price Level
- Customer Type
- Site
- Channel
- Sales Date
- Visit Date

Ticket Funnel

Show:

Sold → Issued → Activated → Used → Expired/Unused

Drill-Down

Example:

Tickets Sold

- Site
- Attraction
- Product
- Ticket Type
- Transaction

AI Capability

AI shall identify unusual ticket-product behavior.

Example:

“Family Pass sales increased 23% following the weekend promotion and now represent 17% of admission revenue.”

Acceptance Criteria

Users can analyze the complete commercial lifecycle of ticket products.

Screen 5.3 — Product & Attraction Performance

Purpose

Identify which products, attractions, events and packages generate the strongest and weakest commercial performance.

Product KPIs

- Units Sold
- Gross Revenue
- Net Revenue
- Revenue Contribution %
- Average Selling Price
- Discount %
- Refund %
- Conversion Rate
- Attendance
- Revenue per Visitor

- Capacity Utilization
- Growth %

Rankings

Display:

Top Selling Products

Highest Revenue Products

Fastest Growing Products

Lowest Performing Products

Highest Refund Products

Highest Conversion Products

Product Performance Matrix

Plot products using:

Revenue × Volume × Growth × Margin/Contribution where available

Low-Performance Identification

The source matrix explicitly identifies products with zero sales over an extended period or inactive products as candidates for review/removal and requires duplicate and low-performing SKUs to be flagged.

AI Capability

AI can recommend:

“Product X generated only 0.4% of revenue during the last six months and overlaps significantly with Product Y. Consider reviewing its catalogue position.”

AI recommendations shall not automatically deactivate products.

Acceptance Criteria

Commercial teams can compare product and attraction performance and identify optimization opportunities.

Screen 5.4 — Sales Channel Performance

Purpose

Compare performance across all TICVAI sales and distribution channels.

Supported Channels

- B2C Web
- Mobile App
- POS
- Kiosk
- B2B
- Corporate
- Reseller
- OTA
- Partner API
- Call Center
- Other configured channels

Channel KPIs

- Gross Sales
- Net Revenue

- Transactions
- Tickets Sold
- Average Order Value
- Average Ticket Price
- Conversion
- Discounts
- Refunds
- Cancellations
- Commission
- Net Contribution
- Growth
- Target Achievement

The source matrix requires sales-channel reporting including ticket types, serial numbers, quantity sold, targets, commissions, refunds, cancellations, discounts and expiry information.

Channel Comparison

Example:

Channel	Revenue	Share	AOV	Refund %	Growth
Online	—	—	—	—	—
POS	—	—	—	—	—
B2B	—	—	—	—	—
OTA	—	—	—	—	—

AI Capability

AI shall identify channel shifts and opportunities.

“Mobile App conversion is 18% higher than Web, but App traffic represents only 14% of digital sessions.”

Acceptance Criteria

Users can compare channel performance using consistent commercial KPIs.

Screen 5.5 — Promotion & Discount Analytics

Purpose

Measure the effectiveness and financial impact of promotions, discounts, vouchers and promotional pricing.

KPI Summary

- Promotion Revenue
- Discount Value
- Discount %
- Orders Using Promotion
- Tickets Sold Under Promotion
- Average Discount
- Incremental Revenue
- Conversion Uplift

- Refunds on Discounted Sales
- Promotion ROI where measurable

Promotion Analysis

Display:

- Promotion Name
- Promotion Code
- Validity Period
- Standard Price
- Promotional Price
- Discount Offered
- Quantity Sold
- Revenue
- Attraction
- Channel
- Date/Time

These fields are specifically required in the source matrix.

Discount Analysis

Analyze by:

- Discount Type
- POS
- Staff
- Ticket Type
- Attraction
- Site
- Channel
- Customer Segment
- Approval Type

The matrix separately requires discount reporting with POS ID, staff details, date/time, ticket type, attraction, location and sales channel.

AI Capability

AI can detect:

- Over-discounting
- Poor-performing promotions
- High-performing promotions
- Unusual staff discount activity
- Cannibalization indicators

Acceptance Criteria

Management can measure promotion effectiveness and identify potentially abnormal discount behavior.

Screen 5.6 — Upsell & Cross-Sell Analytics

Purpose

Measure incremental revenue generated by upselling and cross-selling throughout the customer journey.

KPI Summary

- Upsell Revenue
- Cross-Sell Revenue
- Attachment Rate
- Upsell Conversion
- Cross-Sell Conversion
- Average Incremental Spend
- Commission Generated

Origin Analysis

Track:

Originating Location / Attraction / Business Unit

→

Destination Product / Attraction / Business Unit

The source requirement specifically calls for upsell/cross-sell reporting by originating and destination location/attraction/business unit, date, staff and commission.

Example

Admission Ticket

- F&B Meal
- Photo Package
- Fast Pass
- Retail Merchandise

Staff Performance

Analyze:

- Staff
- POS
- Location
- Upsell Count
- Revenue
- Conversion
- Commission

AI Capability

AI may identify opportunities such as:

“Guests purchasing Family Passes have a 31% F&B attachment rate but only 8% photo-package attachment. Offering the photo package during checkout represents a cross-sell opportunity.”

Acceptance Criteria

Commercial teams can measure incremental sales and determine where cross-sell and upsell strategies are effective.

Screen 5.7 — Refund, Cancellation & Revenue Leakage Analytics

Purpose

Provide management visibility into revenue lost through refunds, cancellations, voids, discounts and other commercial leakage.

KPI Summary

- Refund Value
- Refund Transactions
- Refund Rate
- Cancellation Value
- Cancellation Rate
- Void Value
- Discount Leakage
- Complimentary Value
- Revenue at Risk

Analysis

Analyze by:

- Site
- Product
- Channel
- Cashier
- Staff
- Customer
- Refund Reason
- Cancellation Reason
- Payment Method
- Approval Level

The source matrix requires sales-refund reporting containing ticket number, ticket type and sale/refund dates.

It also specifically requires a control mechanism ensuring commissions are not added on refunds.

Anomaly Detection

AI shall identify patterns such as:

- Abnormally high refund rate
- Repeated refunds by staff member
- Excessive manual discounting
- High void rate at a POS
- Refunds after ticket usage
- Unusual cancellation concentration

Acceptance Criteria

Authorized users can identify commercial leakage and drill down to supporting transactions according to their permissions.

Screen 5.8 — Reservation & Advance Sales Analytics

Purpose

Separate the analysis of **when revenue/sales are generated** from **when the customer will actually visit**.

This is particularly important for attractions, events and advance bookings.

KPIs

- Reservations Created
- Future Reservations
- Booking Value
- Deposits
- Fully Paid Reservations
- Outstanding Balance
- Average Lead Time
- Cancellation Rate
- No-Show Rate
- Advance Booking Revenue

Analysis

Display:

Booking Date vs Visit Date

Booking Window

Analyze:

- Same Day
- 1–3 Days
- 4–7 Days
- 8–30 Days
- 31–90 Days
- 90+ Days

The source matrix explicitly requires reporting where the sales transaction occurs in the current month while reservations are future dated.

It also contains required reports for reservations by date/account/pick-up date, deposits, booking status, group bookings, cancellations, no-shows, booking source and advance bookings.

AI Forecast

AI shall use forward bookings to estimate future:

- Attendance
- Revenue
- Capacity demand
- Expected no-shows

Acceptance Criteria

Users can distinguish booking activity, recognized sales, future attendance and forward commercial demand.

Screen 5.9 — Revenue Recognition & Commercial Allocation

Purpose

Provide analytical visibility into how ticket and commercial revenue moves between sale, consumption, earning and allocation.

Revenue States

Display:

Sold→ **Paid**→ **Deferred/Unearned**→ **Consumed/Used**→ **Earned/Recognized****KPI Summary**

- Gross Sales
- Collected Revenue
- Deferred Revenue
- Earned Revenue
- Unused Ticket Liability
- Breakage
- Allocated Revenue
- Unallocated Revenue

The matrix specifically requires reporting of admission sales with earned and unearned revenue.

Package/Combo Allocation

For bundled products:

Ticket + F&B + Retail + External Product

show:

- Package Selling Price
- Component Value
- Allocation Method
- Revenue Share
- Internal Revenue
- External Partner Revenue

The source requirements explicitly require reports for external combo tickets and the respective split of revenue.

Important Boundary

This screen is **analytics only**.

Actual accounting rules, journals and financial posting remain under the **Finance Module**.

Acceptance Criteria

Commercial and finance users can analyze revenue status and allocation without duplicating financial transaction-processing functions.

Screen 5.10 — Sales Forecast, Targets & AI Commercial Intelligence**Purpose**

Bring historical sales, targets, current performance and AI forecasting together into a commercial decision-support screen.

KPI Summary

- Revenue Target
- Actual Revenue
- Forecast Revenue
- Target Achievement %
- Forecast Variance

- Tickets Target
- Actual Tickets
- Forecast Tickets
- Expected Attendance
- Expected Conversion

Forecasting Dimensions

Forecast by:

- Site
- Attraction
- Product
- Channel
- Day
- Week
- Month

Forecast Inputs

Where available, analytics may consider:

- Historical sales
- Booking pace
- Seasonality
- Day of week
- Event calendar
- Capacity
- Promotions
- Current booking position
- Historical attendance

AI Commercial Insights

AI shall categorize recommendations:

Opportunity

Risk

Trend

Anomaly

Forecast

Example

Opportunity

Saturday afternoon sessions are forecast to reach 96% capacity. Demand is currently 14% above the comparable four-week average.

Risk

B2B revenue is forecast to finish 9% below monthly target based on current booking pace.

Trend

Mobile sales have increased for six consecutive weeks and now represent 22% of digital revenue.

Recommended Actions

Where supported by configuration, AI may suggest:

- Review promotion

- Increase campaign focus
- Review capacity
- Review channel allocation
- Investigate refund pattern
- Evaluate upsell opportunity
- Review dynamic pricing recommendation

Governance

AI shall **recommend**, not directly change pricing, capacity, promotions or commercial configuration unless a separately governed workflow explicitly permits it.

Acceptance Criteria

Management can compare actuals, targets and forecasts and receive explainable AI-supported commercial recommendations.

Board-Wide Filters

All applicable screens should share a consistent global filter framework:

Tenant / Organization

Country

Site

Venue

Attraction

Business Unit

Event

Product

Ticket Type

Channel

Customer Segment

Date / Visit Date

Currency

Filters should persist as the user navigates between Board 5 analytics screens where appropriate.

Board-Wide Drill-Down Principle

A major KPI should not stop at the chart.

For example:

Net Revenue

AED 8.4M

↓

Site

↓

Attraction

↓

Channel

↓

Product

↓

Ticket Type

↓

Transaction

Similarly:

Refund Rate

4.2%

↓

Site

↓

Channel

↓

Product

↓

Staff

↓

Refund Transaction

All drill-down must respect RBAC and tenant/site permissions.

Board-Wide AI Commercial Copilot

Board 5 should include a persistent **Ask TICVAI** capability.

Management could ask:

“Why is ticket revenue below target today?”

“Which products grew the most this month?”

“Which channel has the highest refund rate?”

“Compare online sales against POS for the last eight weekends.”

“Which promotions generated the highest incremental revenue?”

“Where are we losing the most revenue?”

“Forecast ticket sales for next weekend.”

AI should return:

Answer + Visualization + Contributing Factors + Supporting Data + Recommended Action

rather than only conversational text.

Board 5 Screen Summary

#	Screen	Main Purpose
5.1	Sales & Revenue Command Center	Overall commercial performance
5.2	Ticket Sales Analytics	Ticket-product sales and lifecycle
5.3	Product & Attraction Performance	Product/attraction ranking
5.4	Sales Channel Performance	B2C/POS/B2B/OTA/channel analytics
5.5	Promotion & Discount Analytics	Promotion effectiveness and discount control
5.6	Upsell & Cross-Sell Analytics	Incremental commercial revenue
5.7	Refund, Cancellation & Revenue Leakage	Commercial leakage and anomalies

#	Screen	Main Purpose
5.8	Reservation & Advance Sales Analytics	Forward bookings and booking behavior
5.9	Revenue Recognition & Commercial Allocation	Earned/deferred/allocation analytics
5.10	Sales Forecast, Targets & AI Intelligence	Forecasting and commercial recommendations

Board 5 Core Flow

Commercial Overview → Ticket Sales → Product Performance → Channel Performance → Promotions → Upsell/Cross-Sell → Leakage → Forward Bookings → Revenue Allocation → Forecast & AI Actions

This board should give the **Chief Commercial Officer / Sales Director / Revenue Manager / Ticketing Manager** one place to answer the key commercial questions: **What are we selling? Where are we selling it? What is driving growth? Where are we losing revenue? What will happen next? And what should we do about it?**

Board 6 — Finance, Payment & Reconciliation Analytics

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide CFOs, finance teams, revenue accountants, treasury teams, and authorized management users with a unified analytical workspace covering **payments, settlements, reconciliation, receivables, deferred/earned revenue, taxes, cost/profit centers, cashier balancing, and financial exceptions**.



Board 6 shall **not replace the TICVAI Finance Module**. The Finance Module performs financial operations and configuration; Board 6 provides the **BI, monitoring, reconciliation analytics, exception intelligence, and AI explanation layer**.

The matrix specifically requires payment-gateway reconciliation, ERP reconciliation, pending/unreconciled transactions, aging, revenue reporting, cost/profit-center reporting, payment summaries, transaction journals, deferred revenue, refunds, voids and discrepancy reporting.

Screen 6.1 — Finance Analytics Command Center

Purpose

Provide finance management with a consolidated real-time view of financial performance and financial-control status.

KPI Cards

Display:

- Gross Revenue
- Net Revenue
- Payments Collected
- Refunds
- Discounts
- Taxes

- Deferred Revenue
- Recognized Revenue
- Outstanding Receivables
- Payment Gateway Pending
- ERP Unreconciled
- Cash Over/Short
- Reconciliation Rate
- Financial Exceptions

Revenue Breakdown

Analyze by:

Site → Business Unit → Revenue Type → Channel → Product

Business units may include:

- Ticketing
- F&B
- Retail
- Membership
- B2B
- Parking
- Add-ons
- Other configured services

Period Comparison

Support:

- Today vs Yesterday
- MTD vs Previous Month
- YTD vs Previous Year
- Actual vs Budget
- Actual vs Target
- Actual vs Forecast

AI Finance Summary

Example:

“Net revenue is 6.3% above target. AED 184K remains unreconciled, primarily from two payment gateway settlement batches.”

Acceptance Criteria

Finance management can understand financial performance and control exceptions without manually combining multiple reports.

Screen 6.2 — Payment & Tender Analytics

Purpose

Provide complete analytical visibility into how customers paid and how payment methods are performing.

Payment KPIs

- Total Payments
- Successful Payments

- Failed Payments
- Pending Payments
- Refunded Payments
- Payment Success Rate
- Average Payment Value
- Payment Processing Fees

Tender Types

Analyze:

- Cash
- Credit Card
- Debit Card
- Digital Wallet
- Gift Card
- Voucher
- Account/Credit
- Bank Transfer
- Other configured tender

The matrix requires daily transaction reporting by currency amount sold, discounted amount, tender type and units sold.

Payment Analysis

Break down by:

Tender → Provider → Channel → Site → POS → Transaction

Payment Failure Analytics

Display:

- Failure Reason
- Provider
- Channel
- Device/POS
- Time
- Failure %
- Lost Transaction Value

AI Capability

Example:

“Card authorization failures increased from 1.8% to 4.7% between 14:00 and 15:00, primarily affecting Gateway B.”

Acceptance Criteria

Finance and operations teams can analyze tender usage, payment performance and payment failures.

Screen 6.3 — Payment Gateway Reconciliation

Purpose

Compare TICVAI transactions against payment-gateway settlement information.

Three-Way Comparison

TICVAI Transaction

↔

Payment Gateway

↔

Settlement / Bank Record**KPI Summary**

- Transactions to Reconcile
- Automatically Matched
- Partially Matched
- Unmatched
- Missing Settlement
- Amount Variance
- Refund Variance
- Reconciliation %

Reconciliation Status**Matched****Partially Matched****Unmatched****Missing****Under Investigation****Variance Analysis**

Detect:

- Amount mismatch
- Missing transaction
- Duplicate settlement
- Missing refund
- Incorrect fee
- Currency mismatch
- Settlement-date mismatch

The matrix explicitly requires full reconciliation involving the payment gateway, refunds, ticket usage/liability and related financial information.

AI Matching

AI may recommend likely matches where references differ but amount/date/customer/transaction characteristics strongly correlate.

AI recommendations must remain reviewable.

Acceptance Criteria

Finance users can identify and investigate payment-gateway reconciliation differences from transaction to settlement level.

Screen 6.4 — ERP & Financial System Reconciliation**Purpose**

Compare TICVAI commercial/financial transactions against postings transferred to the client's ERP or financial system.

Reconciliation Flow**TICVAI**

→ **Integration Layer**→ **ERP**→ **Posting Confirmation****KPI Summary**

- Transactions Generated
- Transactions Sent to ERP
- Successfully Posted
- Pending Posting
- Failed Posting
- Unreconciled
- Amount Variance
- Posting Success %

The source requirements specifically require reporting for bulk transactions integrated with Finance/ERP and transactions still unreconciled with Finance/ERP.

Analysis**Filter by:**

- Site
- Business Unit
- Posting Date
- Transaction Date
- Batch
- Journal
- Cost Center
- Profit Center
- **Status**

Exception Reasons**Examples:**

- ERP unavailable
- Invalid account mapping
- Missing cost center
- Invalid tax mapping
- Amount discrepancy
- Duplicate posting
- Posting rejected

AI Root-Cause Analysis

“68% of today’s failed ERP postings reference an inactive cost-center mapping introduced in the latest finance configuration.”

Acceptance Criteria

Finance can reconcile TICVAI against ERP postings without relying solely on integration/server logs.

Screen 6.5 — Accounts Receivable & Aging Analytics

Purpose

Provide visibility into outstanding balances and customer/corporate receivables.

KPIs

- Total Receivables
- Current
- Overdue
- Average Collection Period
- Accounts Over Limit
- Advance Payments
- Outstanding Corporate Balance
- Credit Write-Offs

Aging Buckets

Configurable buckets such as:

Current

1–30 Days

31–60 Days

61–90 Days

91–120 Days

120+ Days

Analyze By

- Corporate Account
- B2B Partner
- Customer Account
- Site
- Business Unit
- Invoice
- Currency

The matrix requires account-aging, client-account movement, advance-payment, account-history, billing-account sales, credit write-off and outstanding-balance reports.

AI Capability

AI shall identify:

- Increasing aging exposure
- High-risk accounts
- Unusual balance growth
- Collection deterioration

Acceptance Criteria

Finance teams can identify outstanding receivables and drill down from aging bucket to account and supporting transaction.

Screen 6.6 — Revenue Recognition & Deferred Revenue Analytics

Purpose

Provide finance with analytical visibility into the relationship between **sales date, service/visit date, consumption and revenue recognition**.

Revenue Lifecycle

Sold → Paid → Deferred → Consumed → Recognized

KPI Summary

- Sales Value
- Cash Collected
- Deferred Revenue
- Recognized Revenue
- Outstanding Liability
- Expired/Unused Value
- Breakage
- Future-Dated Revenue

The matrix requires reporting for admissions showing used/unused tickets and earned versus unearned revenue.

Analysis

Break down by:

- Product
- Ticket Type
- Membership
- Package
- Site
- Sale Date
- Visit Date
- Recognition Period

Revenue Movement

Show:

Opening Deferred Revenue

- New Deferred Revenue
- Recognized Revenue
- Refunds/Adjustments

= Closing Deferred Revenue

Important Boundary

This is an **analytics/reconciliation view**.

Accounting recognition rules and journal generation remain under the Finance Module.

Acceptance Criteria

Finance can explain deferred and recognized revenue positions down to the supporting product/transaction level.

Screen 6.7 — Tax, Cost Center & Profit Center Analytics

Purpose

Provide consolidated analysis of taxes and organizational financial allocations.

Tax Analytics

Display:

- Taxable Sales
- Tax-Exempt Sales
- Tax Collected
- Tax Adjustments
- Refund Tax
- Net Tax Liability
- Tax by Site
- Tax by Product
- Tax by Business Unit

The matrix requires tax reports in approved government formats and finance-configurable reporting.

Cost/Profit Center Analytics

Display:

- Revenue by Profit Center
- Revenue by Cost Center
- Allocation %
- Allocated Revenue
- Unallocated Revenue
- Allocation Exceptions

The source requirements explicitly require profit-center/cost-center allocation reporting at attraction sub-level where applicable.

Allocation Drill-Down

Organization

- Site
- Attraction
- Business Unit
- Profit Center / Cost Center
- Transaction

AI Capability

AI shall identify abnormal or missing financial allocations.

Acceptance Criteria

Finance can analyze tax and allocation results while preserving the Finance Module as the system responsible for configuration/posting.

Screen 6.8 — Cashier, POS & Shift Reconciliation Analytics

Purpose

Provide finance and operations teams with analytical visibility into POS, cashier and shift balancing.

KPIs

- Opening Float
- Cash Sales
- Card Sales
- Other Tender

- Refunds
- Voids
- Expected Cash
- Declared Cash
- Actual Cash
- Over/Short
- Shift Status

Analysis

Filter by:

- Site
- Outlet
- POS
- Workstation
- Cashier
- Shift
- Date
- Tender Type

The matrix requires nightly balancing including tender balancing, batch-file exports and exception reporting. It also requires transaction reporting by teller/employee, price overrides and refunds.

Exception Detection

Highlight:

- Cash shortage
- Cash overage
- Excessive voids
- Excessive refunds
- Manual price overrides
- Shift not closed
- Tender mismatch

AI Capability

“POS 14 has recorded cash shortages in four of the last six shifts. Total variance: AED 1,280.”

Acceptance Criteria

Finance users can reconcile cashier/shift performance and identify discrepancies requiring investigation.

Screen 6.9 — Financial Exception & Control Center

Purpose

Create one centralized workspace for financial exceptions requiring investigation or action.

Exception Categories

- Payment mismatch
- Gateway mismatch
- ERP mismatch
- Missing settlement

- Duplicate payment
- Failed refund
- Revenue allocation issue
- Tax mismatch
- Cash over/short
- Unreconciled transaction
- Deferred-revenue issue
- Suspicious financial activity

Exception Record

Each exception shall display:

- Exception ID
- Type
- Severity
- Amount
- Site
- Source
- Transaction
- Detected Date/Time
- Owner
- **Status**
- Aging
- Root Cause
- Recommended Action

Workflow

Detected → Assigned → Investigating → Resolved → Verified → Closed

Financial Materiality

Allow configurable thresholds.

Example:

< AED 100 — Low

AED 100–5,000 — Medium

> AED 5,000 — High

Actual thresholds shall be client configurable.

AI Prioritization

AI may prioritize exceptions based on:

Value × Age × Frequency × Risk × Business Impact

Acceptance Criteria

Financial exceptions can be centrally tracked from detection through resolution with full traceability.

Screen 6.10 — AI Finance Intelligence & Forecast

Purpose

Provide CFOs and finance management with forward-looking financial intelligence rather than only historical reporting.

Forecast KPIs

- Forecast Revenue
- Forecast Cash Collection
- Forecast Refunds
- Forecast Deferred Revenue
- Forecast Recognized Revenue
- Forecast Receivables
- Expected Settlement
- Expected Financial Variance

Forecast Horizons

- Tomorrow
- Next 7 Days
- Month End
- Quarter End
- Custom Period

AI Insight Categories**Revenue****Cash****Reconciliation****Receivables****Risk****Anomaly****Forecast****Example — Forecast**

“Based on current booking pace and historical utilization, September net revenue is forecast at AED 24.8M, approximately 3.7% above target.”

Example — Risk

“AED 620K of corporate receivables will move into the 90+ day aging bucket within the next 14 days if no payment is received.”

Example — Reconciliation

“Gateway A accounts for 74% of unreconciled transactions today. Most exceptions originated between 09:30 and 10:15.”

Ask TICVAI

Finance users should be able to ask:

“Why are receivables increasing?”

“Show unreconciled payments above AED 5,000.”

“Compare recognized revenue with last month.”

“Which sites have the highest refund rate?”

“Explain today's ERP reconciliation variance.”

“Forecast month-end revenue.”

Governance

AI shall provide **analysis and recommendations**, but shall not:

- Post journals
- Write off balances

- Change tax configuration
- Process refunds
- Change accounting mappings
- Close financial exceptions

without an authorized operational workflow and appropriate approval.

Acceptance Criteria

Finance management receives explainable forecasts and recommendations backed by governed TICVAI financial data.

Board-Wide Financial Filters

All applicable screens should provide common filters:

Organization / Tenant

Country

Site / Venue

Business Unit

Attraction

Product

Sales Channel

Payment Method

Payment Gateway

Corporate/B2B Account

Cost Center

Profit Center

Transaction Date

Visit/Service Date

Accounting Period

Currency

Board-Wide Reconciliation Principle

A critical design requirement for Board 6 is the ability to trace financial information end-to-end.

For example:

AED 250 Ticket Sale

Order

↓

Ticket

↓

Payment

↓

Payment Gateway

↓

Settlement

↓

TICVAI Finance Record

↓

ERP Posting

↓

Revenue Recognition

The finance user should be able to identify **where the value currently sits and where any mismatch occurred**.

Board 6 Screen Summary

#	Screen	Main Purpose
6.1	Finance Analytics Command Center	Overall finance performance
6.2	Payment & Tender Analytics	Payment-method performance
6.3	Payment Gateway Reconciliation	Gateway/settlement reconciliation
6.4	ERP & Financial System Reconciliation	TICVAI-to-ERP reconciliation
6.5	Accounts Receivable & Aging Analytics	Receivables and aging
6.6	Revenue Recognition & Deferred Revenue	Earned/unearned revenue analytics
6.7	Tax, Cost Center & Profit Center Analytics	Tax and financial allocation
6.8	Cashier, POS & Shift Reconciliation	Cash/tender/shift balancing
6.9	Financial Exception & Control Center	Manage finance exceptions
6.10	AI Finance Intelligence & Forecast	Forecasting and AI analysis

Board 6 Core Flow

Finance Overview → Payment Analysis → Gateway Reconciliation → ERP Reconciliation → Receivables → Revenue Recognition → Tax & Allocation → Cashier Reconciliation → Exceptions → AI Forecast

The important difference from Board 5 is that **Board 5** answers “How is the business selling?”, while **Board 6** answers “Where is the money, does it reconcile, is it financially correct, and what financial risk needs attention?”.

Board 7 — Operations, Access & Capacity Analytics

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide venue operations, access-control teams, duty managers, attraction managers, and senior management with a **real-time operational intelligence center** covering attendance, entry/exit activity, gate performance, occupancy, capacity utilization, resource utilization, queues, guest flow, and operational exceptions.



This board is the analytical counterpart of the operational modules we already designed. It shall **not configure gates, access rules, capacity rules, virtual queues, resources, or ticket validity**. Instead, it consumes their data and answers:

How many guests are here? Where are they? Are gates operating correctly? Where is capacity under pressure? Where are queues developing? What operational issue needs attention next?

The reporting matrix specifically requires people counts, ticket counts by gate/site, admission usage, entry totals, barcode audits, gate utilization, attendance by hour/day/product and occupancy reporting. It separately requires capacity and resource-utilization analytics, including available-versus-booked capacity, peak utilization and session occupancy.

Screen 7.1 — Live Operations Command Center

Purpose

Provide a real-time consolidated view of operational conditions across all sites.

Live KPI Cards

Display:

- Guests On-Site Now
- Entries Today
- Exits Today
- Current Occupancy
- Occupancy %
- Remaining Capacity

- Active Gates
- Gate Availability %
- Average Queue Time
- Active Sessions
- Resource Utilization %
- Operational Alerts
- Access Rejection Rate

Site Status

Each site shall be classified:

Normal / Warning / Critical

Live Site Table

Display:

Site	On-Site	Capacity	Occupancy	Gates	Queue	Alerts
Adventure Park	—	—	—	—	—	—
Water Park	—	—	—	—	—	—

Operational Map

Where supported, show venues/sites on an interactive map with status indicators.

AI Operations Summary

Example:

“Overall operations are normal. Adventure Park is approaching 91% capacity, while Main Gate rejection rates have increased during the last 20 minutes.”

Acceptance Criteria

A duty manager can understand the current operational state of all authorized sites without opening multiple modules.

Screen 7.2 — Attendance & People Count Analytics

Purpose

Provide detailed analysis of visitor attendance and footfall.

KPIs

- Total Admissions
- Unique Visitors
- Current On-Site
- Entries
- Exits
- Peak Attendance
- Average Attendance
- Repeat Entries
- Re-Entries
- Attendance vs Forecast
- Attendance vs Capacity

Attendance Timeline

Display visitor count by:

- 15 minutes
- 30 minutes
- Hour
- Day
- Week
- Month

The source matrix explicitly requires hourly footfall reporting with peak hours across attractions, F&B and retail areas.

Breakdown

Analyze attendance by:

- Site
- Attraction
- Product
- Ticket Type
- Membership
- Customer Segment
- Gate
- Entry Method
- Time

Comparative Analytics

Compare:

Today vs Yesterday

Today vs Same Weekday

Actual vs Forecast

Actual vs Capacity

AI Capability

“Attendance between 14:00–16:00 is 17% higher than the comparable four-week average.”

Acceptance Criteria

Users can analyze attendance from enterprise level down to individual attraction/gate level.

Screen 7.3 — Gate & Access Performance

Purpose

Provide real-time and historical analytics for gates, turnstiles, handheld readers and other admission devices.

Gate KPIs

- Total Scans
- Valid Scans
- Rejected Scans
- Rejection Rate
- Entries
- Exits
- Re-Entries
- Override Count

- Average Scan Time
- Gate Throughput
- Device Availability
- Offline Events

Gate Status

Show:

Online

Offline

Degraded

Maintenance

High Rejection

Gate Comparison

Rank gates by:

- Throughput
- Rejection Rate
- Availability
- Utilization
- Average Processing Time

Rejection Reasons

Analyze:

- Already Used
- Expired
- Wrong Date
- Wrong Gate
- Invalid Credential
- Anti-Passback
- Capacity Restriction
- Entitlement Missing
- Offline Validation Issue
- Other configured reason

Drill-Down

Rejected Scans

- Site
- Gate
- Reason
- Ticket Type
- Scan Record

AI Capability

“Gate 6 rejection rate is 3.2× higher than the site average. Most failures relate to tickets assigned to tomorrow's visit date.”

Acceptance Criteria

Operations teams can identify gate/access performance issues without accessing device configuration screens.

Screen 7.4 — Occupancy & Capacity Intelligence

Purpose

Provide real-time visibility into consumed, booked and remaining capacity.

KPIs

- Configured Capacity
- Booked Capacity
- Admitted Capacity
- Remaining Capacity
- Utilization %
- Forecast Utilization
- Peak Utilization
- No-Show %
- Overcapacity Risk

Analyze By

- Site
- Venue
- Attraction
- Zone
- Session
- Timeslot
- Resource
- Event

Capacity Visualization

Each operating entity should display:

Available

Booked

Used

Remaining

Thresholds

Example:

Normal: <80%

Warning: 80–94%

Critical: ≥95%

Thresholds shall be configurable.

Forecast

Show expected occupancy for upcoming periods.

Example:

14:00 — 72%

15:00 — 84%

16:00 — 96% 

17:00 — 91%

AI Capability

“Water Park is forecast to reach 96% occupancy at 16:00 based on current bookings and expected walk-up demand.”

Acceptance Criteria

Operations can identify current and future capacity pressure before it becomes an operational issue.

Screen 7.5 — Queue & Guest Flow Analytics

Purpose

Analyze guest movement, waiting time and congestion across physical and virtual queues.

KPIs

- Guests in Queue
- Average Wait Time
- Maximum Wait Time
- Queue Throughput
- Virtual Queue Guests
- Physical Queue Guests
- Abandonment Rate
- Service Rate
- Congestion Alerts

Queue Analysis

Analyze by:

- Attraction
- Entry Point
- Service Point
- Timeslot
- Queue Type
- Hour

Guest Flow

Where data is available, visualize movement:

Entrance → Attraction → F&B → Retail → Attraction → Exit

Heatmap

Display congestion by:

- Venue Zone
- Attraction
- Time

AI Capability

“Roller Coaster queue time is forecast to exceed 35 minutes within the next 20 minutes due to increased arrivals and reduced throughput.”

Recommended Action

AI may suggest:

- Review queue allocation
- Open additional lane

- Redirect guests
- Review resource availability
- Promote alternative attraction

AI shall not automatically alter operational configurations unless separately authorized.

Acceptance Criteria

Users can identify queue pressure and guest-flow bottlenecks from one analytical screen.

Screen 7.6 — Session, Timeslot & Resource Utilization

Purpose

Measure how effectively bookable operational capacity and resources are being used.

KPIs

- Sessions Scheduled
- Sessions Completed
- Available Capacity
- Booked Capacity
- Actual Attendance
- Session Occupancy %
- Resource Utilization %
- No-Shows
- Unused Capacity

The source matrix specifically requires available-versus-booked capacity, peak-utilization analysis, group utilization, session occupancy and resource-utilization dashboards.

Resource Types

Depending on venue configuration:

- Attractions
- Rooms
- Equipment
- Instructors
- Guides
- Vehicles
- Rental Resources
- Activity Resources
- Other bookable resources

Utilization Matrix

Compare:

Available → Reserved → Used → Unused

Underutilization

Identify:

- Low occupancy sessions
- Unused resources
- High no-show sessions
- Capacity imbalance

AI Capability

“The 10:00 session averages only 48% utilization while the 14:00 session averages 94%. Consider reviewing capacity distribution or demand incentives.”

Acceptance Criteria

Management can understand whether operational resources and scheduled capacity are being used efficiently.

Screen 7.7 — Operational Performance & Service Levels

Purpose

Measure operational efficiency against configured service targets.

KPI Categories

Admission

- Entry Throughput
- Scan Time
- Rejection %

Queue

- Average Wait
- Maximum Wait
- Abandonment %

Capacity

- Utilization
- No-Show %

Resource

- Utilization
- Downtime

Operational Availability

- Gate Availability
- Device Availability
- Attraction Availability

SLA / Target Comparison

Each KPI shall display:

Target | Actual | Variance | Status

Trend Analysis

Track deterioration or improvement over time.

Benchmarking

Compare:

- Site vs Site
- Attraction vs Attraction
- Gate vs Gate
- Current vs Historical
- Actual vs Target

AI Capability

AI shall identify the operational factors having the largest impact on guest experience.

Acceptance Criteria

Management can evaluate operational performance using standardized service-level KPIs.

Screen 7.8 — Operational Exception & Incident Intelligence

Purpose

Centralize analytical visibility into abnormal operating conditions.

Exception Categories

- High Occupancy
- Capacity Risk
- Gate Offline
- High Rejection Rate
- Excessive Queue
- Resource Unavailable
- Session Delay
- Device Failure
- Unexpected Attendance
- Access Override Spike
- Operational Downtime

Exception Record

Display:

- Exception ID
- Severity
- Site
- Attraction
- Category
- KPI
- Expected Value
- Actual Value
- Detected Time
- Duration
- Owner
- **Status**
- Impact

Status Workflow

Detected → Acknowledged → Assigned → Investigating → Resolved

Correlation

TICVAI shall correlate related events where possible.

Example:

Gate Offline

→ Reduced throughput

→ Longer entrance queue

→ Increased waiting time

AI Root-Cause Analysis

“Entrance congestion is likely associated with two offline turnstiles, reducing processing capacity by approximately 28%.”

Acceptance Criteria

Users can distinguish isolated alerts from related operational incidents and understand their potential impact.

Screen 7.9 — Site, Attraction & Operational Benchmarking

Purpose

Compare operational performance across the TICVAI portfolio.

Comparison KPIs

- Attendance
- Occupancy
- Capacity Utilization
- Gate Throughput
- Rejection Rate
- Average Queue
- Resource Utilization
- No-Show Rate
- Operational Availability
- Incidents
- Revenue per Visitor where authorized

Rankings

Display:

Highest Utilization

Lowest Utilization

Best Gate Performance

Highest Queue Time

Highest No-Show

Best Operational Improvement

Benchmark Options

Compare:

Site vs Site

Attraction vs Attraction

Current Period vs Previous Period

Actual vs Target

Normalization

Where appropriate, metrics should normalize differences in venue size.

Examples:

- Entries per Gate per Hour
- Revenue per Visitor
- Incidents per 10,000 Guests
- Transactions per POS

- Capacity Utilization %

This prevents a large venue from appearing better/worse simply because it handles more guests.

AI Capability

“Site B has 14% lower throughput per gate than comparable sites despite similar attendance. Longer scan processing time is the primary contributor.”

Acceptance Criteria

Management can benchmark operational efficiency fairly across different sites and attractions.

Screen 7.10 — AI Operations Intelligence & Predictive Control

Purpose

Provide operations teams with forward-looking AI intelligence that identifies emerging problems before they materially affect guests.

AI Insight Categories

Forecast

Congestion Risk

Capacity Risk

Access Anomaly

Resource Risk

Operational Opportunity

Predictive KPIs

Forecast:

- Attendance
- Occupancy
- Queue Time
- Gate Demand
- Session Utilization
- Resource Demand
- No-Shows
- Operational Exceptions

Example — Congestion Risk

High Risk — Main Entrance

Arrival volume is forecast to exceed current gate processing capacity between 15:10 and 15:40.

Cause:

High booking arrival concentration + two unavailable lanes.

Recommendation:

Review lane availability and direct eligible guests toward Entrance B.

Example — Capacity Opportunity

Wildlife Tour's 17:00 session is forecast at 46% utilization while 16:00 is forecast at 97%.

Recommendation:

Consider encouraging eligible guests toward the 17:00 session.

Ask TICVAI

Users may ask:

“Which attraction has the longest queue right now?”

“Why did gate rejection increase?”

“Which site will reach capacity first?”

“Forecast attendance for the next three hours.”

“Which resources are underutilized today?”

“Compare gate performance across all sites.”

Explainability

Every AI recommendation should provide:

Observation → Likely Cause → Business/Guest Impact → Recommendation → Confidence → Supporting Data

Governance

AI shall **recommend operational actions**, but shall not independently:

- Open/close gates
- Override access
- Change capacity
- Modify ticket validity
- Move reservations
- Change virtual-queue rules

unless separately enabled through an approved workflow.

Acceptance Criteria

Operations users receive actionable, explainable forecasts based on authorized real-time and historical operational data.

Board-Wide Real-Time Data Principle

Board 7 is different from many other BI boards because much of its information must be **near-real-time**.

The architecture should therefore distinguish:

Live Operational Data

Examples:

Gate status
Entries/exits
Current occupancy
Queues
Device status

Near-Real-Time Analytics

Examples:

Hourly throughput
Current utilization
Rejection patterns

Historical BI

Examples:

Monthly attendance trends
Site benchmarking
Seasonal utilization

We should **not depend solely on a slow Power BI refresh cycle for live operational monitoring**. TICVAI should have a real-time operational data layer, with Power BI/BI services used where appropriate for analytical visualization.

Board-Wide Operational Filters

Common filters should include:

Organization

Site

Venue

Attraction

Zone

Gate

Session

Timeslot

Resource

Ticket Type

Access Credential Type

Date / Time

Board 7 Screen Summary

#	Screen	Main Purpose
7.1	Live Operations Command Center	Overall real-time venue status
7.2	Attendance & People Count Analytics	Footfall and visitor analytics
7.3	Gate & Access Performance	Gate/reader/access analytics
7.4	Occupancy & Capacity Intelligence	Live and forecast capacity
7.5	Queue & Guest Flow Analytics	Queue and congestion intelligence
7.6	Session, Timeslot & Resource Utilization	Operational-resource efficiency
7.7	Operational Performance & Service Levels	SLA/KPI performance
7.8	Operational Exception & Incident Intelligence	Exceptions and root-cause analysis
7.9	Site, Attraction & Operational Benchmarking	Cross-site comparison
7.10	AI Operations Intelligence & Predictive Control	Predictive operational intelligence

Board 7 Core Flow

Live Operations → Attendance → Gates → Capacity → Guest Flow → Resources → Service Levels → Exceptions → Benchmarking → Predictive AI

This board therefore becomes the **operational brain of the Unified Analytics platform**: Board 5 tells management **how the business is selling**, Board 6 tells Finance **where the money is and whether it reconciles**, and Board 7 tells Operations **what is happening inside the venue right now, what is likely to happen next, and where intervention may be required**.

Board 8 — Customer, CRM, Membership & Loyalty Analytics

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide marketing, CRM, membership, loyalty, commercial, and executive teams with a unified **Customer Intelligence & Engagement Analytics Center** that connects customer identity, visits, purchases, memberships, loyalty activity, campaigns, and behavioral data into one analytical view.



This board shall answer questions such as:

Who are our customers? Where did they come from? How often do they visit? What do they buy? How much are they worth? Which customers are becoming inactive? Which memberships are performing best? How effective is loyalty? Which campaigns actually generate revenue?

Board 8 is strictly the **analytics layer**. Customer profile maintenance remains in CRM; membership configuration remains in Membership; loyalty rules remain in Loyalty; and campaign execution remains in Marketing.

Screen 8.1 — Customer Intelligence Command Center

Purpose

Provide a consolidated management view of customer acquisition, engagement, retention and value.

KPI Cards

Display:

- Total Customers
- Identified Customers
- New Customers
- Returning Customers
- Active Customers
- Repeat Visitor %

- Average Visits per Customer
- Average Spend
- Customer Lifetime Value
- Active Members
- Membership Conversion %
- Loyalty Members
- Loyalty Redemption %
- Customer Retention %
- At-Risk Customers

Customer Trend

Analyze:

New vs Returning vs Reactivated Customers

by:

- Day
- Week
- Month
- Year

Customer Contribution

Display:

- Revenue from New Customers
- Revenue from Returning Customers
- Revenue from Members
- Revenue from Loyalty Members
- Revenue from Guest/Anonymous Customers

AI Customer Summary

Example:

“Returning customers generated 41% of revenue this month, up 6.8% from the previous period. Membership customers have an average spend 23% higher than non-members.”

Acceptance Criteria

Management can understand customer health and commercial contribution from one consolidated analytical screen.

Screen 8.2 — Customer 360 Analytics

Purpose

Provide an analytical 360-degree view of an individual customer or customer segment.

This should complement—not duplicate—the operational CRM Customer 360 screen.

Customer Analytics

Display:

Profile Summary

- Customer ID
- Customer Segment
- Registration Date
- Preferred Site

- Membership Status
- Loyalty Status

Behavior

- Total Visits
- Last Visit
- Visit Frequency
- Average Visit Duration
- Favorite Attraction
- Preferred Visit Day/Time

Commercial

- Lifetime Spend
- Ticket Spend
- F&B Spend
- Retail Spend
- Add-On Spend
- Average Order Value

Engagement

- Campaign Engagement
- Promotion Usage
- Loyalty Activity
- Membership Activity

Customer Journey Timeline

Visualize:

Acquired → First Purchase → First Visit → Repeat Visit → Membership → Loyalty → Recent Engagement

Privacy

Access to personally identifiable information shall depend on the user's permissions.

Where required, analytical users should see anonymized or aggregated data.

Acceptance Criteria

Authorized users can understand the complete relationship between a customer and TICVAI services without manually combining data from multiple modules.

Screen 8.3 — Customer Segmentation & Behavioral Analytics

Purpose

Allow users to analyze customers using demographic, commercial and behavioral characteristics.

Segmentation Dimensions

Support:

- New / Returning
- Visitor Frequency
- Spend Level
- Customer Lifetime Value
- Membership Status
- Loyalty Tier

- Geography
- Acquisition Source
- Preferred Attraction
- Preferred Channel
- Product Preference
- Visit Frequency
- Recency

RFM Analytics

Where configured, TICVAI should support:

Recency

How recently did the customer visit/buy?

Frequency

How often do they visit/buy?

Monetary Value

How much do they spend?

Example Segments

- VIP Customers
- High Value / High Frequency
- High Value / Low Frequency
- Frequent Families
- Weekend Visitors
- Lapsed Customers
- New Digital Customers
- Members Near Renewal
- High Loyalty Engagement
- At-Risk Customers

Segment Comparison

Compare:

Customers | Revenue | Visits | AOV | CLV | Retention

AI Segmentation

AI may identify behavioral clusters automatically.

Example:

“A segment of 8,420 customers visits primarily on weekends, purchases family admission products and has a 34% higher F&B attachment rate.”

Acceptance Criteria

Authorized users can create and analyze customer segments using governed customer information.

Screen 8.4 — Customer Acquisition & Conversion Analytics

Purpose

Understand where customers originate and how successfully acquisition activity converts into customers and revenue.

Acquisition Sources

Analyze:

- Direct
- Organic Search
- Paid Search
- Social
- Email
- Affiliate
- B2B
- OTA
- Partner
- Referral
- Walk-In
- Other configured sources

Funnel

Visualize:

Reach/Traffic

↓

Engagement

↓

Registration

↓

Purchase

↓

Visit

↓

Repeat Visit

↓

Membership

KPIs

- Customers Acquired
- Acquisition Conversion %
- First Purchase Value
- First Visit %
- Acquisition Cost where available
- Revenue per Acquisition Source
- Repeat Rate
- Membership Conversion
- Customer Lifetime Value

Channel Quality

TICVAI should distinguish between channels producing **volume** and channels producing **valuable customers**.

Example:

Source	Customers	Conversion	Avg. Spend	CLV
Social	—	—	—	—
Search	—	—	—	—
B2B	—	—	—	—
Referral	—	—	—	—

AI Capability

“Referral customers represent only 9% of new customers but have 27% higher twelve-month value than paid-social customers.”

Acceptance Criteria

Marketing can understand customer acquisition quality rather than relying only on campaign clicks or traffic.

Screen 8.5 — Retention, Visit Frequency & Churn Intelligence**Purpose**

Measure whether customers continue engaging with venues over time and identify customers likely to become inactive.

KPIs

- Retention Rate
- Repeat Visit Rate
- Average Visit Frequency
- Days Between Visits
- Reactivated Customers
- Dormant Customers
- At-Risk Customers
- Churn Rate
- Returning Revenue

Cohort Analytics

Create cohorts by:

- First Purchase Month
- First Visit Month
- Membership Start
- Acquisition Channel
- Customer Segment

Example:

Month 0 → Month 1 → Month 3 → Month 6 → Month 12

Churn Indicators

AI may consider:

- Reduced visit frequency
- Reduced spending
- Membership expiration

- Loyalty inactivity
- Lower campaign engagement
- Historical behavior patterns

AI Churn Risk

Example:

1,280 high-value customers are showing declining engagement and have a predicted elevated churn risk.

Recommended Actions

AI may suggest:

- Retention campaign
- Membership renewal offer
- Loyalty incentive
- Re-engagement communication

AI shall not automatically send communications without an approved campaign workflow.

Acceptance Criteria

CRM and marketing teams can identify retention patterns and customer segments requiring attention.

Screen 8.6 — Membership Performance & Renewal Analytics

Purpose

Provide comprehensive analytics for memberships and subscriptions.

Membership KPIs

- Active Members
- New Members
- Renewed Members
- Expired Members
- Membership Revenue
- Renewal Rate
- Membership Conversion Rate
- Average Member Spend
- Average Member Visits
- Member Lifetime Value
- Benefits Consumed
- Members Near Expiry

Membership Funnel

Eligible Customer

→ **Membership Offered**

→ **Membership Purchased**

→ **Active**

→ **Renewal Due**

→ **Renewed / Expired**

Membership Comparison

Analyze by:

- Membership Type

- Tier
- Site
- Customer Segment
- Acquisition Channel
- Period

Member vs Non-Member

Compare:

- Spend
- Visit Frequency
- F&B Spend
- Retail Spend
- Retention
- Average Order Value

Renewal Forecast

Show:

- Expiring Next 7 Days
- Expiring Next 30 Days
- Expiring Next 60 Days
- Predicted Renewals
- Renewal Risk

AI Capability

“Gold Members whose visit frequency exceeded four visits during their first 90 days have a significantly higher historical renewal rate.”

Acceptance Criteria

Management can understand membership growth, value, usage and renewal performance.

Screen 8.7 — Loyalty Earn, Burn & Engagement Analytics

Purpose

Measure the effectiveness and financial/behavioral impact of the loyalty program.

Loyalty KPIs

- Active Loyalty Members
- Points Earned
- Points Redeemed
- Earn/Burn Ratio
- Redemption Rate
- Outstanding Points
- Expiring Points
- Loyalty-Driven Revenue
- Active Tiers
- Tier Movement

Earn Analysis

Analyze points earned by:

- Product
- Site
- Channel
- Customer Segment
- Membership Type
- Promotion

Burn Analysis

Analyze redemption by:

- Reward
- Product
- Site
- Channel
- Customer Segment

Loyalty Liability

Where financial values are assigned to points, display:

- Outstanding Points
- Estimated Liability
- Expiring Liability
- Redeemed Liability

Tier Analytics

Display movement:

Entry → Silver → Gold → Platinum

including:

- Upgrades
- Downgrades
- Retention
- Spend by Tier
- Visits by Tier

AI Capability

“Gold-tier members have accumulated AED-equivalent loyalty value but redemption activity is 19% below historical levels.”

Acceptance Criteria

Users can understand whether the loyalty program is driving engagement and repeat commercial behavior.

Screen 8.8 — Campaign & Marketing Performance Analytics

Purpose

Connect marketing activity to actual ticketing, attendance and revenue outcomes.

KPIs

- Campaigns Active
- Audience Reached

- Engagement
- Clicks
- Conversions
- Tickets Sold
- Revenue Generated
- Conversion Rate
- Cost per Conversion
- ROI / ROAS where cost data exists
- Repeat Visits
- Membership Conversions

Campaign Funnel

Audience

→ **Delivered**

→ **Opened/Viewed**

→ **Clicked**

→ **Purchased**

→ **Visited**

→ **Repeated**

Campaign Comparison

Compare by:

- Campaign
- Channel
- Audience Segment
- Product
- Site
- Promotion

Attribution

Where data allows, support configurable attribution approaches such as:

- First Touch
- Last Touch
- Campaign Code
- Promotion Code
- Direct Attribution

The platform should avoid presenting inferred attribution as certain where the available data does not support it.

AI Capability

“Campaign A generated more transactions, but Campaign B produced 31% higher average customer value and stronger repeat visitation.”

Acceptance Criteria

Marketing teams can connect campaign activity to commercial and visitation outcomes.

Screen 8.9 — Customer Value & Commercial Behavior Analytics

Purpose

Understand total customer value across the TICVAI ecosystem rather than ticket spend alone.

Customer Value Components

Ticketing

-

F&B

-

Retail

-

Membership

-

Add-ons

-

Other Services

=

Total Customer Value

KPIs

- Customer Lifetime Value
- Average Spend per Visit
- Ticket Spend
- F&B Spend
- Retail Spend
- Membership Spend
- Add-On Spend
- Purchase Frequency
- Cross-Domain Attachment Rate

Spend Distribution

Segment customers:

- Low Value
- Medium Value
- High Value
- VIP / Top Value

Thresholds shall be configurable.

Basket Behavior

Analyze relationships such as:

Ticket Product → F&B → Retail → Add-On

Cross-Sell Affinity

Example:

Customers buying **Family Admission** are 2.1 × more likely to purchase **Meal Packages** than the general visitor population.

AI Capability

AI can identify:

- High-value segments

- Product affinity
- Cross-sell opportunities
- Customer-value growth
- Declining customer value

Acceptance Criteria

Commercial teams can evaluate the full economic value of customer relationships across authorized TICVAI domains.

Screen 8.10 — AI Customer Intelligence & Next-Best-Action

Purpose

Provide forward-looking AI intelligence that converts customer analytics into explainable recommendations.

AI Insight Categories

Acquisition

Retention

Membership

Loyalty

Customer Value

Campaign

Cross-Sell

Churn Risk

Next-Best-Action Engine

Examples:

Customer Segment: Family Visitors

Observation:

High visit frequency but low membership penetration.

Recommendation:

Evaluate targeted membership conversion campaign.

Customer Segment: Gold Members

Observation:

High loyalty balance with declining redemption.

Recommendation:

Consider a loyalty redemption campaign before point expiry.

Customer Segment: Lapsed High-Value Customers

Observation:

No visit for 120+ days despite historically high engagement.

Recommendation:

Evaluate personalized reactivation campaign.

AI Opportunity Ranking

Prioritize recommendations using:

Potential Value × Probability × Customer Impact × Urgency

Ask TICVAI

Authorized users should be able to ask:

“Which customer segment has the highest lifetime value?”

“Why is membership renewal decreasing?”

“Which campaigns generated the highest revenue?”

“Which customers are most likely to churn?”

“Compare member and non-member spending.”

“Which customer segment should we target for membership?”

“What is driving loyalty redemption?”

AI Response

Return:

Answer

- **Visualization**
- **Supporting Data**
- **Reasoning Factors**
- **Recommended Action**
- **Confidence**

Governance

AI shall not automatically:

- Send campaigns
- Change customer segments
- Award/remove loyalty points
- Upgrade membership
- Change membership pricing
- Modify customer consent
- Trigger personalized communication

without an authorized workflow and appropriate permissions.

Acceptance Criteria

AI recommendations are explainable, permission-aware and connected to supporting analytical evidence.

Board-Wide Customer Privacy & Security

This is especially important for Board 8.

Analytics shall respect:

- Tenant isolation
- Site permissions
- Role-based access
- Customer consent where applicable
- PII permissions
- Data masking
- Retention rules
- Export restrictions
- Audit logging

An executive may need:

Customer Segment + Revenue + Behavior

without needing:

Name + Email + Mobile Number

Therefore aggregated/anonymized analysis should be the default where individual identity is unnecessary.

Board-Wide Customer Filters

Common filters should include:

Organization

Site

Venue

Attraction

Customer Segment

Membership Type

Loyalty Tier

Acquisition Source

Marketing Channel

Campaign

Product

Visit Frequency

Spend Tier

Date / Period

Important Cross-Domain Architecture

Board 8 becomes powerful only if TICVAI links customer identity across modules.

Conceptually:

Customer ID

↓

Ticket Purchase

↓

Visit / Access

↓

F&B Purchase

↓

Retail Purchase

↓

Membership

↓

Loyalty

↓

Campaign Engagement

This creates a governed **Customer 360 analytical model**.

Without this unified customer identity, Power BI would simply display separate CRM, Ticketing, F&B and Membership reports rather than true customer intelligence.

Board 8 Screen Summary

#	Screen	Main Purpose
8.1	Customer Intelligence Command Center	Overall customer performance
8.2	Customer 360 Analytics	Unified customer analytical view
8.3	Customer Segmentation & Behavioral Analytics	Behavioral segmentation
8.4	Customer Acquisition & Conversion Analytics	Acquisition quality and funnel
8.5	Retention, Visit Frequency & Churn Intelligence	Retention and churn
8.6	Membership Performance & Renewal Analytics	Membership intelligence
8.7	Loyalty Earn, Burn & Engagement Analytics	Loyalty performance
8.8	Campaign & Marketing Performance Analytics	Campaign-to-revenue analytics
8.9	Customer Value & Commercial Behavior Analytics	CLV and cross-domain spending
8.10	AI Customer Intelligence & Next-Best-Action	Predictive recommendations

Board 8 Core Flow

Customer Overview → Customer 360 → Segmentation → Acquisition → Retention → Membership → Loyalty → Campaigns → Customer Value → AI Next-Best-Action

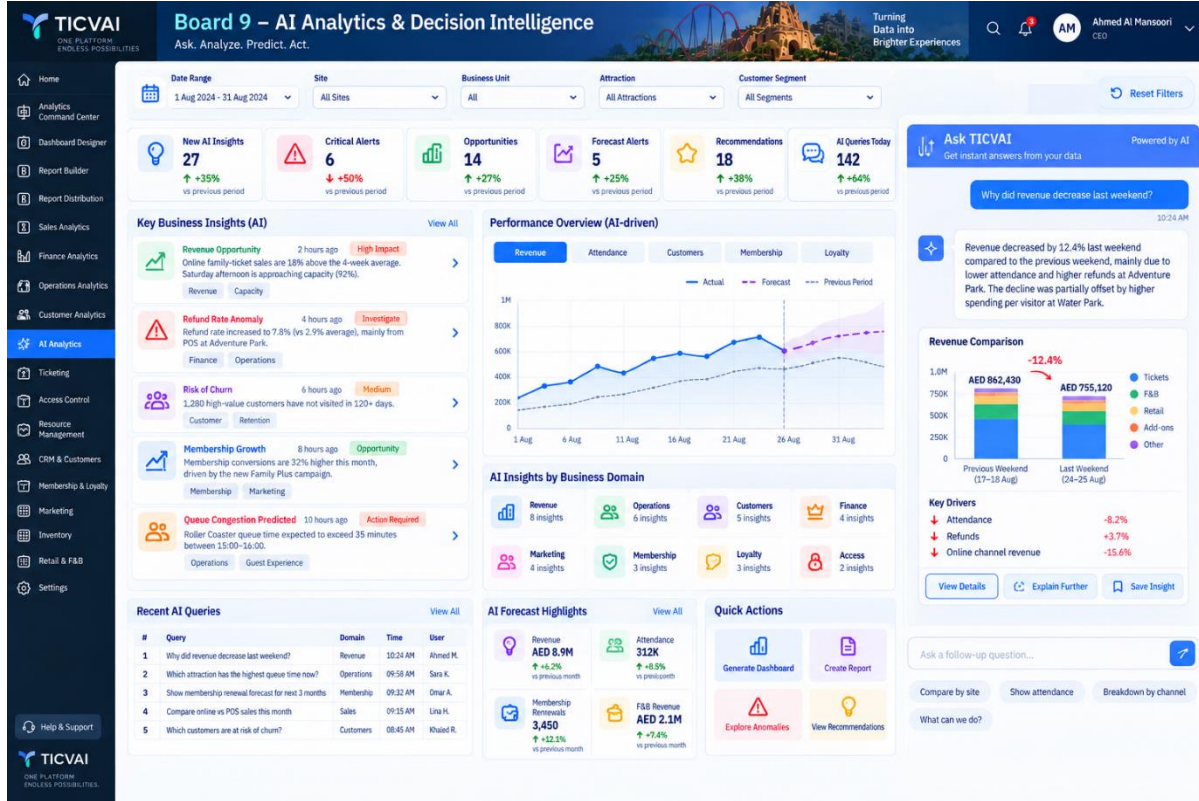
This board should give TICVAI something beyond conventional CRM reporting: a **unified customer intelligence layer** connecting **who the customer is, what they buy, where they visit, how frequently they return, what membership/loyalty relationship they have, how valuable they are, and what the business should do next.**

Board 9 — AI Analytics, Natural Language BI & Decision Intelligence

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide TICVAI with a centralized **AI-powered analytics and decision-intelligence layer** that allows authorized users to ask business questions in natural language, automatically generate dashboards and reports, discover anomalies and trends, understand root causes, generate forecasts, and receive explainable recommended actions.



Board 9 is not another static reporting board. It is the **AI intelligence layer sitting above Boards 1–8**.

It should allow a CEO, CFO, Operations Director, Commercial Director, Marketing Manager, or authorized analyst to ask:

“Why did revenue decrease yesterday?”

and TICVAI should determine the relevant datasets, analyze the information, generate the appropriate visualization, explain the likely drivers, and recommend actions.

This directly addresses the matrix requirements for **AI analytics dashboards, natural-language reporting, AI-generated dashboards, trend analysis, forecasting, real-time/historical analytics and cross-domain reporting**.

Screen 9.1 — AI Analytics Command Center

Purpose

Provide a centralized landing page for all AI-generated analytical intelligence across TICVAI.

AI Insight Categories

Display:

- Executive Insights
- Revenue Insights
- Operations Insights
- Customer Insights
- Finance Insights

- Marketing Insights
- Access Insights
- Membership/Loyalty Insights
- Forecasts
- Anomalies
- Opportunities
- Risks

KPI Summary

Show:

- New AI Insights
- Critical Insights
- Opportunities Detected
- Risks Detected
- Active Anomalies
- Forecast Alerts
- Recommendations Pending Review
- AI Queries Today

Priority Feed

Each insight shall contain:

Insight

Business Domain

Severity / Opportunity

Potential Impact

Confidence

Detected Time

Recommended Action

Example

Revenue Opportunity

Online family-ticket demand is 18% above the comparable four-week average while Saturday afternoon availability is approaching capacity.

Potential Impact: High

Confidence: 91%

Recommendation: Review capacity and dynamic-pricing options.

Acceptance Criteria

Management can review prioritized AI intelligence across all authorized business domains from one screen.

Screen 9.2 — Ask TICVAI — Natural Language Analytics

Purpose

Allow users to query TICVAI business information using normal language instead of manually building reports.

Query Interface

Provide a conversational interface:

Ask TICVAI anything about your business...

Examples:

“How much revenue did we generate yesterday?”

“Compare online sales against POS this month.”

“Which attraction has the highest refund rate?”

“Why did attendance decrease last weekend?”

“Which customers are most likely to renew?”

“Which gates have the highest rejection rate?”

Response Structure

TICVAI should return:

Direct Answer

-

KPI / Visualization

-

Supporting Data

-

Comparison

-

AI Explanation

-

Suggested Follow-Up Questions**Follow-Up Conversation**

Users shall be able to continue contextually:

“Break that down by site.”

then:

“Only show Dubai.”

then:

“Compare it against last year.”

The AI should retain the analytical context within the session.

Acceptance Criteria

Authorized business users can obtain governed analytics without knowing report names, database fields, SQL, or Power BI.

Screen 9.3 — AI-Generated Dashboard Studio**Purpose**

Allow users to generate dashboards from natural-language requests.

Example Request

“Create a dashboard for the Operations Director showing attendance, capacity, queue times, gate rejection and operational incidents.”

AI Shall Propose

- Dashboard Name
- Relevant KPIs
- Datasets
- Visualizations

- **Filters**
- Layout
- Drill paths
- Refresh frequency

Generated Dashboard Preview

Users can:

Accept

Modify

Regenerate

Add Widget

Remove Widget

Save Draft

Send for Approval

Example AI Selection

For the request above:

- Current Attendance → KPI
- Occupancy → Gauge
- Attendance Trend → Line Chart
- Queue Time → Heatmap
- Gate Rejection → Bar Chart
- Operational Incidents → Table
- AI Summary → Insight Card

Governance

AI-generated dashboards shall enter the same lifecycle defined in Board 2:

Draft → Validate → Review → Approve → Publish

AI must not directly publish unrestricted dashboards.

Acceptance Criteria

Users can convert natural-language analytical requirements into governed dashboard configurations.

Screen 9.4 — AI Report Generator

Purpose

Generate detailed reports from natural-language instructions.

Example

User asks:

“Create a monthly report showing ticket sales by site, attraction, product and channel, including gross revenue, discounts, refunds and net revenue.”

AI Determines

Domain: Ticketing + Sales

Dimensions:

- Site
- Attraction
- Product

- Channel

Measures:

- Tickets
- Gross Revenue
- Discount
- Refund
- Net Revenue

Period: Monthly**Report Preview**

Show:

- Selected Dataset
- Fields
- **Filters**
- Grouping
- Calculations
- Sort
- Output

User Actions**Edit Fields****Change Filter****Add Calculation****Save****Run****Schedule****Relationship to Board 3**

Board 3 provides the manual **Self-Service Report Builder**.

Board 9 allows AI to configure that same governed report engine automatically.

Acceptance Criteria

AI-generated reports use the same approved semantic model, permissions, calculations and governance rules as manually created reports.

Screen 9.5 — Anomaly Detection Center**Purpose**

Automatically identify unusual behavior across TICVAI data without requiring users to manually monitor every KPI.

Anomaly Categories

Detect abnormal patterns involving:

- Revenue
- Sales
- Refunds
- Discounts
- Payments
- Attendance

- Gate Rejections
- Capacity
- Queue Times
- Membership
- Loyalty
- Customer Behavior
- F&B
- Retail
- Inventory
- Resources

Each Anomaly Displays

- KPI
- Expected Range
- Actual Value
- Variance
- Severity
- Start Time
- Duration
- Affected Site
- Affected Domain
- Confidence

Example

High Refund Anomaly

Expected Refund Rate: **2.1–3.0%**

Actual: **7.8%**

Affected Channel: **POS**

Affected Site: **Adventure Park**

Detection Methods

Depending on the KPI and available data:

- Configured threshold
- Historical baseline
- Statistical deviation
- Seasonal baseline
- Peer comparison
- AI/model-based detection

Acceptance Criteria

TICVAI can automatically surface material deviations requiring investigation.

Screen 9.6 — Root-Cause Analysis Explorer

Purpose

Help users understand **why** a KPI changed.

This is one of the most important AI capabilities in the entire platform.

Example

User selects:

Revenue ↓ 12.4%

AI investigates contributing dimensions such as:

- Site
- Product
- Channel
- Attendance
- Price
- Discounts
- Refunds
- Capacity
- Customer Segment
- Campaign
- Time

Root-Cause Tree

Example:

Revenue –12.4%

↓

Attendance –8.2%

-

Refunds +3.7%

↓

Primary Drivers:

Site B –AED 184K

Online Channel –AED 92K

Family Pass –AED 48K

Contribution Analysis

Display:

Driver	Contribution	Confidence
Lower attendance	54%	High
Higher refunds	21%	High
Channel mix	14%	Medium
Other factors	11%	—

Important AI Principle

TICVAI should distinguish between:

Correlation

and

Confirmed Cause

AI must not present a statistical relationship as proven causation without sufficient evidence.

Acceptance Criteria

Users can move from detecting a KPI change to understanding the strongest supported contributing factors.

Screen 9.7 — Forecasting & Predictive Analytics Studio

Purpose

Provide configurable forward-looking analytics across business domains.

Forecast Areas

Support forecasts for:

- Revenue
- Ticket Sales
- Attendance
- Occupancy
- Capacity
- Queue Demand
- Refunds
- Cash Collection
- Membership Renewals
- Customer Churn
- F&B Demand
- Retail Demand
- Resource Utilization

Forecast Horizon

Configure:

- Next Hour
- Today
- Tomorrow
- Next 7 Days
- Month End
- Quarter
- Custom

Forecast Visualization

Display:

Historical Actual

→

Current Actual

→

Forecast

with confidence range.

Forecast Comparison

Compare:

Forecast vs Target

Forecast vs Budget

Forecast vs Previous Period

Model Information

Authorized technical/analytical users should be able to see:

- Model
- Training/analysis period
- Last refreshed
- Accuracy
- Confidence
- Data quality
- Key contributing variables

Acceptance Criteria

TICVAI provides transparent forecasts rather than unexplained predictive numbers.

Screen 9.8 — AI Recommendation & Next-Best-Action Center

Purpose

Convert analytical insights into prioritized business recommendations.

Recommendation Categories

- Revenue Optimization
- Capacity Optimization
- Customer Retention
- Membership Conversion
- Loyalty Engagement
- Operational Improvement
- Queue Optimization
- Cost/Risk Reduction
- Promotion Optimization
- Cross-Sell/Upsell

Recommendation Structure

Every recommendation should contain:

Observation

Why It Matters

Potential Impact

Recommended Action

Confidence

Supporting Evidence

Example

Membership Opportunity

Observation:

12,450 non-members have visited three or more times during the last 90 days.

Recommendation:

Consider targeting this segment with a membership-conversion campaign.

Potential Audience:

12,450

Potential Value:

Calculated estimate where sufficient data exists.

Recommendation Status

New → Reviewed → Accepted → Rejected → Actioned → Measured

Human Feedback

Users may indicate:

- Useful
- Not Useful
- Incorrect
- Already Known
- Actioned

This feedback should support future AI quality monitoring.

Acceptance Criteria

Recommendations are prioritized, explainable and measurable rather than presented as generic AI suggestions.

Screen 9.9 — AI Insight History, Evidence & Explainability

Purpose

Create a full governance trail for AI-generated analytics and recommendations.

AI Insight Record

Store:

- Insight ID
- User / System Trigger
- Query
- Date/Time
- Business Domain
- Data Sources
- Reporting Period
- **Filters**
 - AI Response
 - Visualization
 - Confidence
 - Recommendation
 - User Action
 - Feedback

Evidence Panel

Users should be able to select:

View Supporting Data

and see the metrics and datasets used to generate the insight.

Explainability

Where applicable, show:

Why did AI produce this insight?

Example:

Revenue risk was generated because booking pace is 14% below the four-week baseline, while remaining days in the sales period are insufficient to meet the current target under the baseline forecast.

Reproducibility

Where data has not materially changed, authorized users should be able to reproduce the analytical query from the stored configuration.

Audit

Capture:

- Who asked
- What data was accessed
- What answer was generated
- Whether it was exported/shared
- Whether recommendation was accepted

Acceptance Criteria

AI analytics remain auditable and traceable to supporting TICVAI data.

Screen 9.10 — AI Analytics Governance & Model Control**Purpose**

Provide administrators with centralized control over how AI may access and use TICVAI analytics.

This screen is critical because Board 9 should **not give an LLM unrestricted access to the TICVAI database**.

AI Data Permissions

Configure allowed domains:

- Ticketing
- Finance
- Customer
- Access
- Operations
- Membership
- Loyalty
- F&B
- Retail
- Inventory

Security Controls

AI shall inherit:

- Tenant Security
- RBAC
- Site Permissions
- Data-Level Security
- PII Restrictions
- Field Masking
- Export Restrictions

If a manager cannot access Site B manually, asking:

“Show me Site B revenue”

must not allow AI to bypass that restriction.

AI Feature Controls

Administrators can enable/disable:

- Natural Language Query
- AI Dashboard Generation
- AI Report Generation
- Forecasting
- Anomaly Detection
- Recommendations
- Customer-Level AI
- Financial AI

Model/Provider Configuration

Architecture should support approved AI providers/models without hard -coding the user experience to one vendor.

Display:

- AI Service
- Model
- Use Case
- **Status**
- Consumption
- Cost
- Response Time
- Error Rate

AI Consumption Monitoring

Track:

- Queries
- Tokens/Units
- Cost
- User
- Tenant
- Module
- Use Case

This will also help TICVAI later implement **AI usage-based commercial charging**.

Guardrails

Configure:

- Maximum data scope
- Sensitive-data restrictions
- AI response logging
- Confidence threshold
- Human approval requirement
- Cost limits
- Usage quotas

Acceptance Criteria

TICVAI administrators retain centralized security, cost, usage and governance control over AI analytics.

Critical AI Architecture Principle

I strongly recommend putting this explicitly into the SOW for Soft Labs:

The AI analytics layer shall not directly query unrestricted production databases. AI requests shall be resolved through TICVAI's governed semantic/analytics layer, enforcing tenant isolation, RBAC, data-level permissions, approved metrics, business definitions and audit controls.

Conceptually:

User

↓

Ask TICVAI

↓

AI Orchestration Layer

↓

Permission & Governance Layer

↓

TICVAI Semantic Model

↓

Analytics / Data Platform

↓

Authorized Result

↓

AI Explanation + Visualization + Recommendation

This is important because otherwise different AI queries could calculate something like **“Net Revenue”** differently from the official BI dashboard.

Board-Wide AI Response Standard

Whenever practical, AI answers should follow a consistent structure:

1. Answer

What happened?

2. Visualization

Show it.

3. Explanation

Why did it happen?

4. Evidence

Which data supports the conclusion?

5. Forecast

What is likely to happen next?

6. Recommendation

What should the user consider doing?

7. Confidence

How certain is the analysis?

This gives TICVAI a much stronger AI experience than simply placing a chatbot beside Power BI.

Board 9 Screen Summary

#	Screen	Main Purpose
9.1	AI Analytics Command Center	Central AI intelligence feed
9.2	Ask TICVAI — Natural Language Analytics	Conversational BI
9.3	AI-Generated Dashboard Studio	Generate dashboards using AI
9.4	AI Report Generator	Generate detailed reports
9.5	Anomaly Detection Center	Detect unusual business behavior
9.6	Root-Cause Analysis Explorer	Explain KPI changes
9.7	Forecasting & Predictive Analytics Studio	Predict future performance
9.8	AI Recommendation & Next-Best-Action Center	Convert insights into actions
9.9	AI Insight History, Evidence & Explainability	AI audit and evidence
9.10	AI Analytics Governance & Model Control	Security, models, usage and cost

Board 9 Core Flow

AI Command Center → Ask TICVAI → Generate Dashboard/Report → Detect Anomaly → Explain Root Cause → Forecast → Recommend Action → Measure/Review → Audit → Govern

This is the board that should make TICVAI's reporting solution feel genuinely **AI-first** rather than “**Power BI with an AI chat box.**” The intelligence layer understands the authorized business model across ticketing, finance, operations, customers and the other TICVAI domains, while the underlying BI/data layer remains the trusted source for calculations and evidence.

Board 10 — KPI, Benchmarking, Forecasting & BI Administration

Scope of Work — Unified BI, Reporting & AI Analytics Platform

Board Objective:

Provide TICVAI administrators, BI administrators, data owners, and authorized business owners with the **central administration and governance layer for the entire Unified BI, Reporting & AI Analytics Platform**.



Boards 1–9 cover the business-facing analytics experience. **Board 10 controls the foundation underneath them:** enterprise KPI definitions, targets, benchmarks, semantic models, data sources, refresh pipelines, data quality, embedded BI configuration, performance, security, and overall analytics platform health.

This board answers:

Where does the data come from? Which KPI definition is official? Is the data fresh and trustworthy? How are sites benchmarked? Are BI services healthy? Who owns each metric?

The reporting requirements call for KPI monitoring, real-time and historical analytics, trend analysis, forecasting, benchmark reporting, embedded analytics, cross-domain reporting and governed reporting capabilities. Board 10 provides the administration required to make those capabilities reliable and consistent.

Screen 10.1 — BI & Analytics Administration Command Center

Purpose

Provide administrators with one consolidated view of the health and governance of the TICVAI analytics platform.

Platform KPI Cards

Display:

- Active Dashboards
- Active Reports
- Published KPIs
- Connected Data Sources
- Active Datasets

- Data Refresh Success %
- Data Quality Score
- Failed Refreshes
- BI Service Health
- AI Analytics Health
- Active Alerts
- Analytics Users

Platform Components

Show status for:

Operational Data

Analytics Warehouse/Lakehouse

Semantic Model

BI Engine

Embedded Analytics

Reporting Engine

AI Analytics Layer

API/Data Export

Status

Each component shall show:

Healthy / Warning / Critical / Offline

AI Administration Summary

Example:

“Overall analytics health is 98.7%. The Finance semantic model is delayed by 18 minutes due to an incomplete ERP refresh. Three dashboards are currently displaying stale Finance data.”

Acceptance Criteria

Administrators can determine overall analytics-platform health without reviewing multiple technical systems.

Screen 10.2 — Enterprise KPI Library

Purpose

Maintain the authoritative catalogue of approved TICVAI business KPIs.

This is essential because every dashboard must use the **same definition** of a metric.

KPI Record

Configure:

- KPI Name
- KPI Code
- Description
- Business Domain
- Business Owner
- Data Owner
- Formula
- Numerator
- Denominator

- Unit
- Currency
- Aggregation
- Source Dataset
- Effective Date
- Version
- **Status**

Example

KPI: Revenue per Visitor

Formula:

Recognized Net Revenue ÷ Unique Admitted Visitors

Owner:

Finance / Commercial

KPI Categories

- Executive
- Sales
- Revenue
- Finance
- Operations
- Access
- Customer
- Membership
- Loyalty
- Marketing
- F&B
- Retail
- Inventory
- Resources

KPI Status

Draft → Review → Approved → Active → Superseded → Retired

Impact Analysis

Before changing a KPI, show:

Used by 14 dashboards

Used by 27 reports

Used by 3 AI models

Used by 6 alerts

Acceptance Criteria

TICVAI maintains one governed definition for each enterprise KPI.

Screen 10.3 — KPI Targets, Thresholds & Scorecards

Purpose

Centrally configure performance targets and thresholds used throughout TICVAI analytics.

Target Configuration

Targets may be configured by:

- Organization
- Site
- Venue
- Attraction
- Business Unit
- Product
- Channel
- Period

Target Types

- Fixed Target
- Budget Target
- Percentage Target
- Historical Benchmark
- Dynamic Target

Status Rules

Example:

Revenue Target: AED 1M

≥100% → On Target

90–99% → Watch

75–89% → Warning

<75% → Critical

Effective Dates

Targets shall support:

- Start Date
- End Date
- Seasonal Targets
- Event-Specific Targets
- Day-of-Week Targets

Scorecards

Create management scorecards containing multiple KPIs.

Example:

Venue General Manager Scorecard

Revenue

Attendance

Capacity

Customer Satisfaction

Queue Time

Revenue per Visitor

Membership Conversion

Acceptance Criteria

Business targets can be managed centrally and reused consistently across dashboards, alerts and AI analytics.

Screen 10.4 — Benchmark & Comparative Analytics Configuration

Purpose

Define how TICVAI compares performance between sites, periods and peer groups.

Benchmark Types

Support:

Actual vs Target

Actual vs Budget

Current vs Previous Period

Current vs Previous Year

Site vs Site

Attraction vs Attraction

Peer Group vs Peer Group

Portfolio Average

Peer Groups

Administrators may define comparable entities.

Example:

Large Theme Parks

- Site A
- Site B
- Site C

or:

Indoor Attractions

- Attraction A
- Attraction B
- Attraction C

Normalized Benchmarks

Support metrics such as:

- Revenue per Visitor
- Transactions per 1,000 Visitors
- Entries per Gate per Hour
- Incidents per 10,000 Visitors
- Revenue per m² where applicable
- Revenue per Operating Hour
- Utilization %

Ranking

Configure:

Top Performer

Above Benchmark

Average

Below Benchmark

Bottom Performer

Acceptance Criteria

TICVAI can benchmark entities fairly using centrally governed comparison rules.

Screen 10.5 — Data Source & Integration Registry

Purpose

Maintain a centralized catalogue of data sources feeding the analytics platform.

TICVAI Sources

Examples:

- Ticketing
- POS
- Access Control
- Finance
- CRM
- Membership
- Loyalty
- F&B
- Retail
- Inventory
- Resources
- Virtual Queue
- Accreditation

External Sources

Examples:

- ERP
- Payment Gateways
- CRM Platform
- Marketing Platforms
- External Access Systems
- Third-Party Operators
- Data Warehouse
- Approved external APIs

Data Source Record

Display:

- Source Name
- Source Type
- Owner
- Connection
- Authentication Method
- Refresh Type
- Last Sync
- Records Processed

- **Status**
- Data Classification

Connection Status

Connected

Delayed

Failed

Disabled

Important Security Principle

Credentials and secrets shall be securely stored and never displayed in clear text.

Acceptance Criteria

Administrators can identify every source feeding BI and understand its current status and ownership.

Screen 10.6 — Semantic Model & Business Data Catalogue

Purpose

Create the governed business layer between raw data and dashboards/AI.

This is one of the most important technical screens in Board 10.

Business Entities

Expose business-friendly entities such as:

Customer

Order

Ticket

Admission

Payment

Product

Site

Membership

Loyalty Transaction

F&B Sale

rather than raw database tables.

Relationships

Visualize:

Customer

↓

Order

↓

Ticket

↓

Admission

and:

Order → Payment

Field Catalogue

For each field:

- Business Name

- Technical Source
- Description
- Data Type
- Domain
- Sensitivity
- Allowed Uses
- Owner

Approved Measures

Examples:

- Net Revenue
- Tickets Sold
- Unique Visitors
- Average Spend
- Refund Rate
- Occupancy

Relationship Validation

Detect:

- Missing relationships
- Circular relationships
- Duplicate aggregation risk
- Invalid cardinality
- Broken references

AI Usage

Only approved semantic fields and measures should normally be exposed to **Ask TICVAI**.

Acceptance Criteria

Dashboards, reports and AI operate against a consistent governed business-data model rather than independently querying raw tables.

Screen 10.7 — Data Refresh, Pipeline & Data Health Monitor

Purpose

Monitor movement of information from operational systems into the analytics platform.

Pipeline Flow

Source

- Ingestion
- Transformation
- Analytics Store
- Semantic Model
- Dashboard / AI

KPI Cards

- Pipelines Running
- Successful Refreshes
- Failed Refreshes

- Average Refresh Duration
- Data Latency
- Stale Datasets
- Records Processed

Refresh Modes

Support:

- Real-Time / Streaming
- Near Real-Time
- Scheduled
- On-Demand

Data Health

Monitor:

- Completeness
- Timeliness
- Accuracy checks
- Duplicate records
- Missing values
- Schema changes
- Reconciliation status

Example Alert

Ticketing Dataset Delayed

Expected Refresh: 10:00

Last Successful Refresh: 09:32

Current Delay: 28 minutes

Affected:

6 dashboards

12 reports

2 AI insights

Acceptance Criteria

Administrators can identify stale or failed analytics data before users unknowingly make decisions from outdated information.

Screen 10.8 — Embedded BI, Workspace & Tenant Administration

Purpose

Configure how BI technology is embedded inside TICVAI and separated across clients/tenants.

Configuration

Manage:

- Tenant
- BI Workspace
- Dataset
- Dashboard
- Report
- Embedded Application

- Environment
- Capacity
- Authentication
- **Status**

Environment Separation

Support:

Development

Testing/UAT

Production

Tenant Isolation

Each client's analytics shall be securely separated.

Example:

Client A

cannot access:

Client B datasets, reports, dashboards or AI results

Embedded Experience

Configure:

- **Navigation**
- Theme
- Branding
- Language
- Default Dashboard
- Mobile behavior
- Full-screen mode

Technology Independence

TICVAI should expose an abstraction layer so that the user experience is not tightly coupled to one BI provider.

Power BI may be the initial engine, but the architecture should allow approved alternatives.

Acceptance Criteria

BI content can be securely embedded and governed across TICVAI tenants and environments.

Screen 10.9 — Analytics Performance, Usage & Cost Monitor

Purpose

Monitor BI adoption, system performance, capacity consumption and analytical operating costs.

Usage KPIs

- Active Analytics Users
- Dashboard Views
- Report Runs
- AI Queries
- Exports
- API Requests
- Peak Concurrent Users

Performance KPIs

- Average Dashboard Load Time
- Query Duration
- Slowest Reports
- Failed Queries
- Dataset Size
- Refresh Duration
- Embedded BI Availability

Usage Analytics

Identify:

- Most Used Dashboards
- Least Used Dashboards
- Most Used Reports
- Unused Reports
- Most Queried KPIs
- Most Active Users/Roles

Cost Analytics

Where supported, monitor:

- BI Capacity Consumption
- Compute Usage
- Storage
- AI Consumption
- AI Cost
- Export Processing
- Tenant Consumption

Tenant Usage

This becomes particularly useful for TICVAI SaaS commercial models:

Tenant	BI Usage	AI Usage	Storage	Status
Client A	—	—	—	—
Client B	—	—	—	—

AI Optimization

“Dashboard X accounts for 18% of BI compute usage but only 2% of dashboard views. Consider optimizing its dataset/query design.”

Acceptance Criteria

TICVAI can understand analytics adoption, performance and cost at platform and tenant level.

Screen 10.10 — Analytics Governance, Security & Audit Center**Purpose**

Provide final governance over the entire BI and analytics environment.

Governance Areas

Monitor:

- Dashboard Permissions
- Report Permissions
- Dataset Permissions
- KPI Changes
- Data Exports
- API Access
- AI Access
- Sensitive Data Usage
- Configuration Changes
- Administrative Actions

Sensitive Data Controls

Classify fields:

Public

Internal

Confidential

PII

Financial Sensitive

Restricted

Governance Policies

Configure:

- Data Masking
- Export Restrictions
- Retention
- Sharing
- AI Access
- API Access
- Approval Requirements

Audit Trail

Capture:

Who

Did What

When

To Which Object/Data

From Where

Result

Governance Exceptions

Examples:

- Unauthorized export attempt
- Excessive data extraction
- Failed permission check
- Sensitive data accessed
- AI request blocked

- Cross-tenant access attempt
- Unapproved KPI modification

AI Governance Monitoring

Show:

- AI Queries
- Blocked Queries
- Sensitive-Data Requests
- AI Cost
- Model Errors
- Low-Confidence Answers
- User Feedback

Acceptance Criteria

TICVAI provides complete traceability and centralized governance over BI, reporting and AI analytics.

Critical Data Architecture for the Complete Module

Board 10 should make the underlying architecture clear to Soft Labs.

The desired structure is:

TICVAI Operational Modules + External Systems

↓

Unified Data Integration Layer

↓

Analytics Data Platform

Warehouse / Lakehouse / Real-Time Store as appropriate

↓

TICVAI Semantic & KPI Layer

↓

BI / Power BI / Embedded Analytics

-

TICVAI AI Analytics Layer

↓

TICVAI User Experience

Dashboards | Reports | Alerts | Ask TICVAI | Forecasts | AI Recommendations

This architecture means that **Power BI is a component of TICVAI's analytics platform—not the owner of TICVAI's business logic.**

One Very Important Requirement for Soft Labs

I would explicitly include this in the SOW:

Business KPI definitions, semantic models, tenant security, data permissions and AI governance shall be maintained within a governed TICVAI analytics architecture and shall not exist exclusively inside individual Power BI reports or dashboards.

Why?

If **Net Revenue** is calculated independently in 20 Power BI reports, eventually we may have 20 slightly different calculations.

Instead:

TICVAI Enterprise KPI**NET_REVENUE**

↓

used by:

Board 1 Executive Dashboard

Board 5 Sales Analytics

Board 6 Finance Analytics

Board 9 Ask TICVAI

Board 10 Benchmarking

All return the **same governed number**.**Board 10 Screen Summary**

#	Screen	Main Purpose
10.1	BI & Analytics Administration Command Center	Platform administration and health
10.2	Enterprise KPI Library	Govern official KPI definitions
10.3	KPI Targets, Thresholds & Scorecards	Targets and performance rules
10.4	Benchmark & Comparative Analytics Configuration	Benchmarking rules
10.5	Data Source & Integration Registry	Manage analytics sources
10.6	Semantic Model & Business Data Catalogue	Govern analytics data model
10.7	Data Refresh, Pipeline & Data Health Monitor	Monitor data pipelines
10.8	Embedded BI, Workspace & Tenant Administration	BI embedding and multi-tenancy
10.9	Analytics Performance, Usage & Cost Monitor	Performance, adoption and cost
10.10	Analytics Governance, Security & Audit Center	Security and governance

Board 10 Core Flow

Platform Health → KPI Governance → Targets → Benchmarks → Data Sources → Semantic Model → Data Health → Embedded BI → Usage & Cost → Security & Audit

With this, the **10-board Unified BI, Reporting & AI Analytics module is complete**:

Board 1: Unified Analytics Command Center

Board 2: Dashboard Designer & KPI Studio

Board 3: Self-Service Report Builder

Board 4: Report Distribution, Scheduling & Governance

Board 5: Sales, Ticketing & Revenue Analytics

Board 6: Finance, Payment & Reconciliation Analytics

Board 7: Operations, Access & Capacity Analytics

Board 8: Customer, CRM, Membership & Loyalty Analytics

Board 9: AI Analytics & Decision Intelligence

Board 10: KPI, Benchmarking, Data & BI Administration

Together, these boards cover the original **8.7 BI & Reporting requirements** while turning them into a much stronger TICVAI architecture: **one governed analytics platform spanning operational BI, self-service reporting, embedded Power BI, real-time monitoring, cross-domain analytics, forecasting and AI-driven decision intelligence**.